PROFESSIONAL HANDBOOK

# Kaggle for ML Engineers

## Competitive Systems & Applied Architecture

A Systems-First Handbook for Mid→Senior Practitioners

12 Core Parts | Reproducible Code | 2024-2025 Edition

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**About the Author**

**Amer Hussein** is a machine learning engineer with extensive experience in competitive data science and production ML systems. This handbook distills patterns from 2024-2025 competition winners, Grandmaster insights, and real-world production lessons into actionable systems for serious practitioners.

## Executive Summary

Kaggle competitions have evolved from academic exercises to a proving ground for production-ready machine learning systems. The gap between competition success and production impact is narrowing, but significant differences remain in objectives, constraints, and evaluation criteria. This handbook provides a systems-first approach to competitive ML engineering, focusing on reproducible patterns that transfer to production environments.

The 2024-2025 competitive landscape reveals several key trends: gradient-boosted decision trees (LightGBM, XGBoost, CatBoost) continue to dominate tabular competitions, with ensembles of 50-100+ models becoming standard for top finishes. AutoML tools like AutoGluon have democratized access to sophisticated stacking architectures, while GPU acceleration through RAPIDS cuML has transformed experimentation velocity. In NLP, decoder-only LLMs (Llama, Mistral, Qwen) are increasingly prevalent, often fine-tuned with LoRA for efficiency.

This handbook covers twelve interconnected domains: production mapping, mental models, feature engineering, model architectures, engineering discipline, ensembling systems, competitive psychology, advanced tactics, production reality, winner analysis, career leverage, and a comprehensive 90-day mastery plan. Each section includes runnable Python code, benchmark data, failure mode analysis, and explicit transfer guidance for production contexts.

**Key Takeaway:** The practitioners who succeed both on Kaggle and in production share a common trait—they treat machine learning as a systems engineering discipline, not a model-selection exercise. This handbook provides the

architectural patterns, code templates, and decision frameworks to build that discipline.

# PART 1 — Strategy & Positioning: Kaggle ↔ Production Mapping

> *Kaggle and production ML share technical foundations but differ fundamentally in objectives, constraints, and success metrics. Understanding the transfer matrix—the techniques that generalize versus those that don't—is essential for practitioners who want to excel in both domains.*

**Key Points:**

- Kaggle optimizes for a single metric; production balances multiple objectives (latency, cost, interpretability)
- Static datasets vs. continuous data flows require different validation and monitoring strategies
- Ensemble techniques that win competitions often face deployment challenges
- Feature engineering patterns transfer well; overfitting strategies do not

## 1.1 The Transfer Matrix

The following matrix maps Kaggle techniques to their production transferability:

**Table 1 Kaggle-to-Production Transfer Matrix**

| Technique | Kaggle Value | Production Transfer | Adaptation Required |
|---|---|---|---|
| Cross-validation discipline | High | Direct | None |
| Feature engineering | High | Direct | Feature store integration |

| | | | |
|---|---|---|---|
| GBDT models (LGBM/XGB/CAT) | High | Direct | Model versioning, A/B testing |
| Deep ensembles (50+ models) | Very High | Limited | Model distillation, latency constraints |
| Stacking architectures | High | Moderate | Simplify to 2-3 levels max |
| Pseudo-labeling | Medium | Limited | Label quality monitoring |
| Test-time augmentation | Medium | Limited | Latency tradeoff analysis |
| Leakage exploitation | High (short-term) | None | N/A - avoid entirely |

## 1.2 Pipeline Comparison

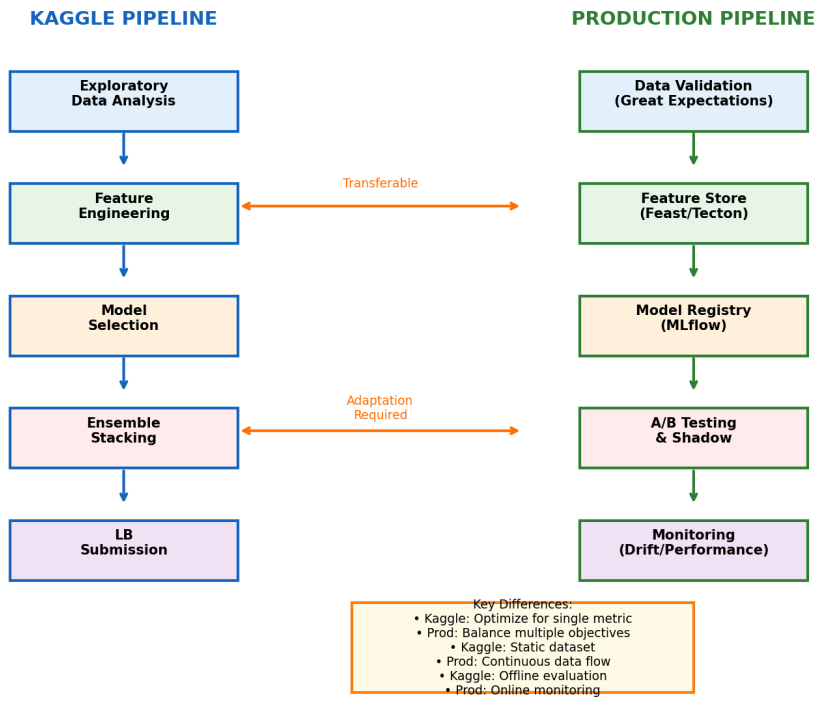


**Figure 1:** Kaggle vs Production Pipeline Comparison

The pipeline comparison reveals structural differences at each stage:

### Feature Engineering Stage

**Kaggle:** Rapid iteration, thousands of features, manual selection
**Production:** Feature store (Feast/Tecton), versioned transformations, online/offline consistency

### Model Management Stage

**Kaggle:** Notebook-based, manual tracking, submission files
**Production:** MLflow registry, automated retraining, canary deployments

## 1.3 What Transfers, What Doesn't

**High-Transfer Patterns (Use in Both Domains)**

- **Rigorous CV design:** Time-aware splits, group preservation, stratification
- **Feature engineering discipline:** Domain-informed transformations, interaction terms
- **Model diversity:** Different algorithms, different features, different seeds
- **Systematic experimentation:** Controlled changes, metric tracking, reproducibility

**Low-Transfer Patterns (Kaggle-Only)**

- **Massive ensembles:** 100+ models impractical for inference
- **Test-set overfitting:** Leaderboard probing, submission-based optimization
- **Leakage exploitation:** Using future information, ID-based features
- **Extreme hyperparameter tuning:** Overfitting to validation set

### Tradeoff Analysis: Ensemble Size

| Ensemble Size | Kaggle LB Gain | Production Latency | Recommendation |
|---|---|---|---|
| 1-3 models | Baseline | Low | Production default |

| | | | |
|---|---|---|---|
| 5-10 models | +1-3% | Moderate | Production with distillation target |
| 20-50 models | +3-5% | High | Kaggle only; distill for production |
| 50+ models | +5-8% | Very High | Kaggle only; requires distillation |

## Failure Mode: The "LB Overfitting" Trap

**Scenario:** A practitioner achieves top-10 on public LB but drops to top-50 on private LB.

**Root Cause:** Hyperparameters tuned to public test set through iterative submission.

**Mitigation:** Use nested CV for hyperparameter selection; never use test set for model decisions.

**Sources & Latest Evidence**

- NVIDIA Grandmasters Playbook (2025-09-18): developer.nvidia.com/blog/the-kaggle-grandmasters-playbook
- State of ML Competitions 2024 (2025-02-25): mlcontests.com/state-of-machine-learning-competitions-2024
- Kaggle Playground Series S5 Analysis (2025-12-05): medium.com/@gauurab/kaggle-playground

# PART 2 — Mental Models: Leaderboard Dynamics & CV Architecture

> ***TL;DR:*** *Your local validation (CV) is your only reliable guide. The leaderboard (LB) is a noisy, potentially biased signal. Build CV architectures that correlate strongly with the final evaluation, detect leakage early, and resist the temptation to overfit to public LB.*

**Key Points:**

- CV-LB correlation > 0.95 is the target; anything below 0.9 signals a problem
- Time-based splits, group-aware folds, and stratification are non-negotiable
- Leakage detection should be automated and run on every new dataset
- Public LB shake-up (public vs private) typically ranges from 5-20% of rankings

## 2.1 Leaderboard Dynamics

Understanding leaderboard behavior is critical for competitive success:

### The LB Signal Model

Think of the public LB as:

```
Public LB Score = True Performance + Random Noise + Potential Bias
```

The noise component depends on test set size:

- Small test sets (n < 10,000): High variance, unreliable for model selection
- Medium test sets (10K-100K): Moderate variance, usable with caution
- Large test sets (n > 100K): Lower variance, more reliable signal

### Shake-up Analysis

Historical shake-up data from 2024 competitions:

**Table 2 Leaderboard Shake-up Patterns (2024 Competitions)**

| Competition Type | Avg Shake-up | Top 10 Stability | Key Factor |
|---|---|---|---|
| Tabular (small data) | 15-25% | 60% | Overfitting variance |
| Tabular (large data) | 5-10% | 85% | Consistent signal |
| NLP | 10-20% | 70% | Distribution shift |
| Computer Vision | 8-15% | 75% | Data augmentation variance |

## 2.2 CV Architecture Patterns

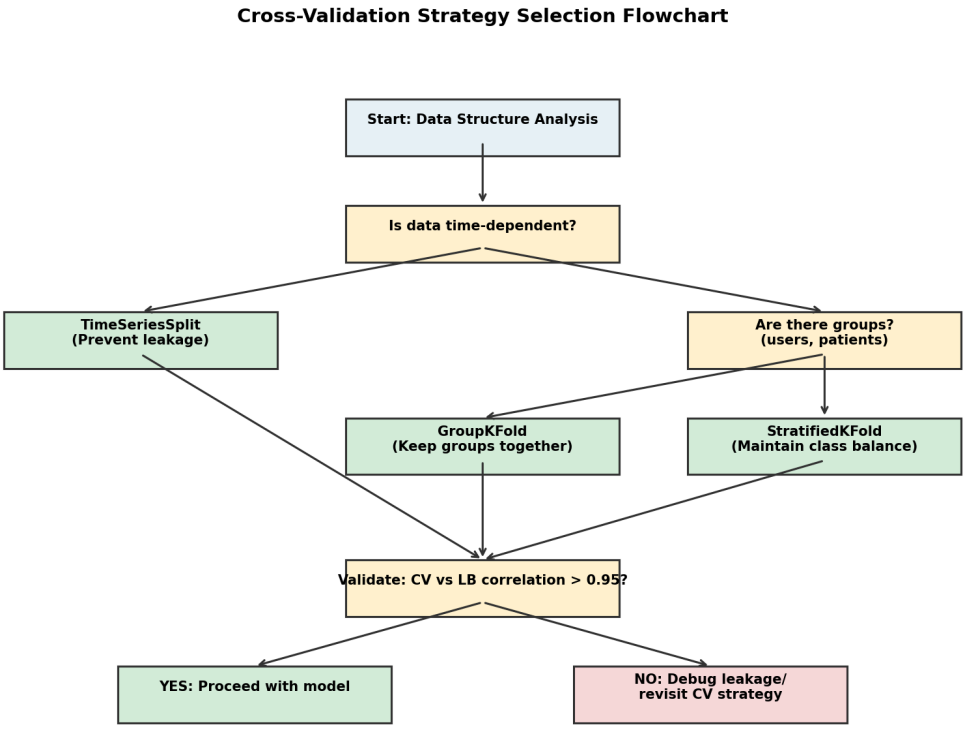


**Figure 2:** Cross-Validation Strategy Selection Flowchart

## Implementation: Robust CV Framework

```
import numpy as np
import pandas as pd
from sklearn.model_selection import (
    KFold, StratifiedKFold, GroupKFold, TimeSeriesSplit
)

def get_cv_strategy(df, target_col, group_col=None, date_col=None,
                    n_splits=5, random_state=42):
    """
    Select appropriate CV strategy based on data characteristics.

    Parameters:
    -----------
    df : pd.DataFrame
        Training data
    target_col : str
        Name of target column
    group_col : str, optional
        Column defining groups (users, patients, etc.)
    date_col : str, optional
        Column with datetime information
    n_splits : int
        Number of CV folds
    random_state : int
        Random seed for reproducibility

    Returns:
    --------
    cv : sklearn CV splitter
        Configured cross-validation object
    """
    y = df[target_col]

    # Priority 1: Time-based data
    if date_col is not None:
        print(f"Using TimeSeriesSplit (date_col={date_col})")
        return TimeSeriesSplit(n_splits=n_splits)

    # Priority 2: Grouped data
    if group_col is not None:
        print(f"Using GroupKFold (group_col={group_col})")
        groups = df[group_col]
        return GroupKFold(n_splits=n_splits)
```

```
import numpy as np
import pandas as pd
from sklearn.model_selection import (
    KFold, StratifiedKFold, GroupKFold, TimeSeriesSplit
)

def get_cv_strategy(df, target_col, group_col=None, date_col=None,
                    n_splits=5, random_state=42):
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        Column defining groups (users, patients, etc.)
    date_col : str, optional
        Column with datetime information
    n_splits : int
        Number of CV folds
    random_state : int
        Random seed for reproducibility

    Returns:
    --------
    cv : sklearn CV splitter
        Configured cross-validation object
    """
    y = df[target_col]

    # Priority 1: Time-based data
    if date_col is not None:
        print(f"Using TimeSeriesSplit (date_col={date_col})")
        return TimeSeriesSplit(n_splits=n_splits)

    # Priority 2: Grouped data
    if group_col is not None:
        print(f"Using GroupKFold (group_col={group_col})")
        groups = df[group_col]
        return GroupKFold(n_splits=n_splits)

    # Priority 3: Classification with imbalance
    if y.nunique() <= 10:
```

### CV-LB Correlation Validation

```
def validate_cv_lb_correlation(cv_scores, lb_scores):
    """
    Calculate correlation between CV and LB scores.
    Target: > 0.95 for reliable validation.
    """
    from scipy.stats import pearsonr, spearmanr

    pearson_r, pearson_p = pearsonr(cv_scores, lb_scores)
    spearman_r, spearman_p = spearmanr(cv_scores, lb_scores)

    print(f"Pearson r: {pearson_r:.4f} (p={pearson_p:.4f})")
    print(f"Spearman r: {spearman_r:.4f} (p={spearman_p:.4f})")

    if pearson_r < 0.9:
        print("WARNING: CV-LB correlation below 0.9")
        print("  - CV may not reflect true performance")
        print("  - Consider revising CV strategy")
    elif pearson_r > 0.95:
        print("CV-LB correlation is strong. CV is reliable.")

    return pearson_r, spearman_r
```

## 2.3 Leakage Detection

Leakage is the silent killer of competition performance. Automated detection is essential.

## Leakage Detection Framework

```
import pandas as pd
import numpy as np
from scipy.stats import ks_2samp, chi2_contingency

class LeakageDetector:
    """
    Automated leakage detection for tabular datasets.
    """

    def __init__(self, train_df, test_df, target_col):
        self.train = train_df.copy()
        self.test = test_df.copy()
        self.target_col = target_col
        self.leakage_report = []

    def check_id_leakage(self, id_cols):
        """Detect if IDs overlap between train and test."""
        for col in id_cols:
            train_ids = set(self.train[col].unique())
            test_ids = set(self.test[col].unique())
            overlap = train_ids.intersection(test_ids)

            if len(overlap) > 0:
                self.leakage_report.append({
                    'type': 'ID Overlap',
                    'column': col,
                    'severity': 'CRITICAL',
                    'details': f"{len(overlap)} IDs appear in both sets"
                })
        return self

    def check_target_leakage(self, feature_cols, threshold=0.99):
        """Detect features perfectly correlated with target."""
        for col in feature_cols:
            if col == self.target_col:
                continue

            # Check for perfect correlation
            correlation = self.train[col].corr(self.train[self.target_col])
            if abs(correlation) > threshold:
                self.leakage_report.append({
                    'type': 'Target Leakage',
                    'column': col,
                    'severity': 'CRITICAL',
```

```
import pandas as pd
import numpy as np
from scipy.stats import ks_2samp, chi2_contingency

class LeakageDetector:
    """
    Automated leakage detection for tabular datasets.
    """

    def __init__(self, train_df, test_df, target_col):
        self.train = train_df.copy()
        self.test = test_df.copy()
        self.target_col = target_col
        self.leakage_report = []

    def check_id_leakage(self, id_cols):
        """Detect if IDs overlap between train and test."""
        for col in id_cols:
            train_ids = set(self.train[col].unique())
            test_ids = set(self.test[col].unique())
            overlap = train_ids.intersection(test_ids)

            if len(overlap) > 0:
                self.leakage_report.append({
                    'type': 'ID Overlap',
                    'column': col,
                    'severity': 'CRITICAL',
                    'details': f"{len(overlap)} IDs appear in both sets"
                })
        return self

    def check_target_leakage(self, feature_cols, threshold=0.99):
        """Detect features perfectly correlated with target."""
        for col in feature_cols:
            if col == self.target_col:
                continue

            # Check for perfect correlation
            correlation = self.train[col].corr(self.train[self.target_col])
            if abs(correlation) > threshold:
                self.leakage_report.append({
                    'type': 'Target Leakage',
                    'column': col,
                    'severity': 'CRITICAL',
                    'details': f"Correlation with target: {correlation:.4f}"
                })
```

## Common Leakage Patterns

**Table 3 Common Leakage Patterns and Mitigations**

| Pattern | Detection | Mitigation | Impact if Missed |
|---|---|---|---|
| ID overlap | Set intersection | Remove ID features | Severe overfitting |
| Target encoding leakage | CV score >> LB score | Use OOF encoding only | Complete model failure |
| Temporal leakage | Future dates in test | Time-based CV | Overoptimistic CV |
| Group leakage | Same group in train/test | GroupKFold | Data contamination |

**Sources & Latest Evidence**

- Scikit-learn CV Documentation: scikit-learn.org/stable/modules/cross_validation
- Kaggle Discussion: Cross-Validation Strategies (2024): kaggle.com/discussions/getting-started/585253
- Enefit Competition Winner Analysis (2024-06-11): linkedin.com/pulse/machine-learning-time-series

# PART 3 — Feature Engineering at Scale

> ***TL;DR:*** *Feature engineering remains the highest-ROI activity in competitive ML. Modern approaches combine automated feature generation with domain-informed selection. The key is building safe, reproducible pipelines that prevent leakage while maximizing signal extraction.*

**Key Points:**

- Winning solutions routinely use 500-2000+ engineered features
- Groupby aggregations are the single most important technique for tabular data
- Target encoding must use out-of-fold (OOF) predictions to prevent leakage
- GPU acceleration (cuDF, Polars) enables feature engineering at scale

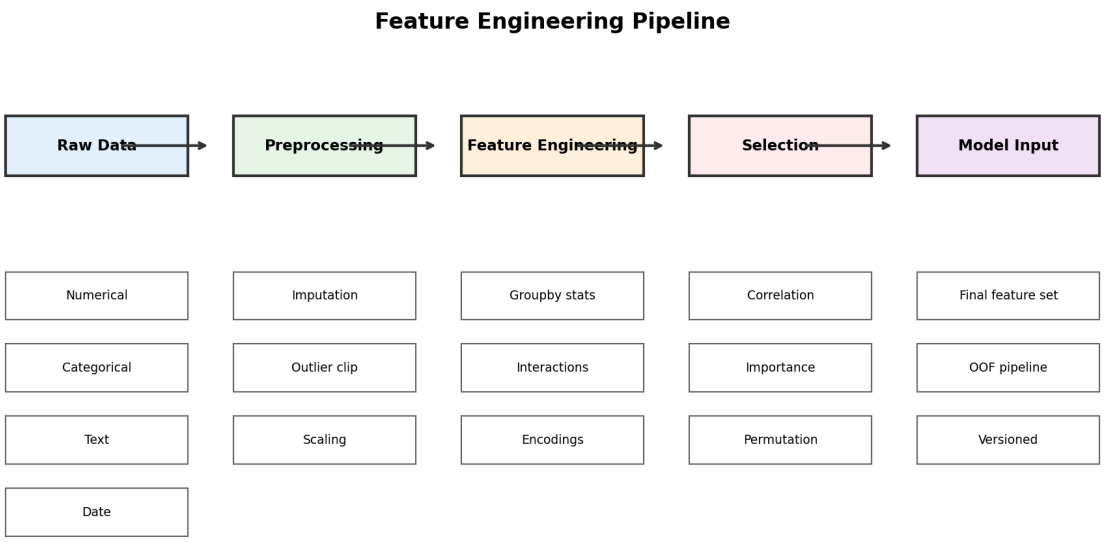


**Figure 3:** Feature Engineering Pipeline Architecture

## 3.1 Feature Stores & Safe Encodings

## Safe Target Encoding with OOF

```
import numpy as np
import pandas as pd
from sklearn.model_selection import KFold

class SafeTargetEncoder:
    """
    Target encoder with out-of-fold predictions to prevent leakage.

    Critical for competition success - never use global target means
    without OOF protection.
    """

    def __init__(self, smoothing=1.0, min_samples_leaf=1, noise=0):
        self.smoothing = smoothing
        self.min_samples_leaf = min_samples_leaf
        self.noise = noise
        self.global_mean = None
        self.encoding_maps = {}

    def fit_transform(self, df, cols, target, n_folds=5, random_state=42):
        """
        Create OOF target encodings using cross-validation.
        """
        df = df.copy()
        self.global_mean = df[target].mean()

        kf = KFold(n_splits=n_folds, shuffle=True, random_state=random_state)

        for col in cols:
            encoded = np.zeros(len(df))

            for train_idx, val_idx in kf.split(df):
                # Calculate means from training fold only
                train_fold = df.iloc[train_idx]
                means = train_fold.groupby(col)[target].agg(['mean', 'count'])

                # Apply smoothing
                smoothed_means = (
                               (means['count'] * means['mean'] + self.smoothing *
self.global_mean) /
                    (means['count'] + self.smoothing)
                )

                # Map to validation fold
```

```
import numpy as np
import pandas as pd
from sklearn.model_selection import KFold

class SafeTargetEncoder:
    """
    Target encoder with out-of-fold predictions to prevent leakage.

    Critical for competition success - never use global target means
    without OOF protection.
    """

    def __init__(self, smoothing=1.0, min_samples_leaf=1, noise=0):
        self.smoothing = smoothing
        self.min_samples_leaf = min_samples_leaf
        self.noise = noise
        self.global_mean = None
        self.encoding_maps = {}

    def fit_transform(self, df, cols, target, n_folds=5, random_state=42):
        """
        Create OOF target encodings using cross-validation.
        """
        df = df.copy()
        self.global_mean = df[target].mean()

        kf = KFold(n_splits=n_folds, shuffle=True, random_state=random_state)

        for col in cols:
            encoded = np.zeros(len(df))

            for train_idx, val_idx in kf.split(df):
                # Calculate means from training fold only
                train_fold = df.iloc[train_idx]
                means = train_fold.groupby(col)[target].agg(['mean', 'count'])

                # Apply smoothing
                smoothed_means = (
                               (means['count']  *  means['mean']  +  self.smoothing  *
self.global_mean) /
                    (means['count'] + self.smoothing)
                )

                # Map to validation fold
                val_fold = df.iloc[val_idx]
                                                                  encoded[val_idx]    =
```

## Groupby Feature Engineering

```
import pandas as pd
import numpy as np

def create_groupby_features(df, group_cols, agg_cols, aggregations=None):
    """
    Create comprehensive groupby aggregation features.

    This is the single most important feature engineering technique
    for tabular competitions.
    """
    if aggregations is None:
        aggregations = ['mean', 'std', 'min', 'max', 'median', 'count', 'nunique']

    df = df.copy()

    for group_col in group_cols:
        for agg_col in agg_cols:
            if agg_col == group_col:
                continue

            # Calculate aggregations
            agg_df = df.groupby(group_col)[agg_col].agg(aggregations)

            # Rename columns
                     agg_df.columns  =  [f'{group_col}_{agg_col}_{agg}'  for  agg  in
aggregations]

            # Merge back
            df = df.merge(agg_df, left_on=group_col, right_index=True, how='left')

            # Create difference/ratio features
            for agg in ['mean', 'median']:
                if f'{group_col}_{agg_col}_{agg}' in df.columns:
                    df[f'{agg_col}_diff_{group_col}_{agg}'] = (
                        df[agg_col] - df[f'{group_col}_{agg_col}_{agg}']
                    )
                    df[f'{agg_col}_ratio_{group_col}_{agg}'] = (
                        df[agg_col] / (df[f'{group_col}_{agg_col}_{agg}'] + 1e-8)
                    )

    return df


# Advanced: Multiple groupby levels
```

```
import pandas as pd
import numpy as np

def create_groupby_features(df, group_cols, agg_cols, aggregations=None):
    """
    Create comprehensive groupby aggregation features.

    This is the single most important feature engineering technique
    for tabular competitions.
    """
    if aggregations is None:
        aggregations = ['mean', 'std', 'min', 'max', 'median', 'count', 'nunique']

    df = df.copy()

    for group_col in group_cols:
        for agg_col in agg_cols:
            if agg_col == group_col:
                continue

            # Calculate aggregations
            agg_df = df.groupby(group_col)[agg_col].agg(aggregations)

            # Rename columns
                     agg_df.columns  =  [f'{group_col}_{agg_col}_{agg}'  for  agg  in
aggregations]

            # Merge back
            df = df.merge(agg_df, left_on=group_col, right_index=True, how='left')

            # Create difference/ratio features
            for agg in ['mean', 'median']:
                if f'{group_col}_{agg_col}_{agg}' in df.columns:
                    df[f'{agg_col}_diff_{group_col}_{agg}'] = (
                        df[agg_col] - df[f'{group_col}_{agg_col}_{agg}']
                    )
                    df[f'{agg_col}_ratio_{group_col}_{agg}'] = (
                        df[agg_col] / (df[f'{group_col}_{agg_col}_{agg}'] + 1e-8)
                    )

    return df


# Advanced: Multiple groupby levels
def create_hierarchical_groupby(df, hier_cols, target_col):
    """
```

## 3.2 OOF Pipelines

Out-of-fold (OOF) predictions are essential for both feature engineering and ensembling. They prevent leakage while enabling powerful meta-features.

```
import numpy as np
from sklearn.base import clone

class OOFPipeline:
    """
    Generic OOF prediction pipeline for any sklearn-compatible model.
    """

    def __init__(self, model, cv):
        self.model = model
        self.cv = cv
        self.oof_predictions = None
        self.models = []

    def fit_predict(self, X, y, return_models=True):
        """
        Generate OOF predictions using cross-validation.
        """
        self.oof_predictions = np.zeros(len(y))
        self.models = []

        for fold, (train_idx, val_idx) in enumerate(self.cv.split(X, y)):
            print(f"Training fold {fold + 1}...")

            X_train, X_val = X.iloc[train_idx], X.iloc[val_idx]
            y_train, y_val = y.iloc[train_idx], y.iloc[val_idx]

            model = clone(self.model)
            model.fit(X_train, y_train)

            # Store OOF predictions
            if hasattr(model, 'predict_proba'):
                self.oof_predictions[val_idx] = model.predict_proba(X_val)[:, 1]
            else:
                self.oof_predictions[val_idx] = model.predict(X_val)

            if return_models:
                self.models.append(model)

        return self.oof_predictions

    def predict(self, X_test):
        """
        Average predictions from all fold models.
        """
        predictions = []
```

## 3.3 Automated Feature Engineering

### FeatureTools Integration

```
import featuretools as ft

def auto_feature_engineering(df, entity_id, target_entity,
                             max_depth=2, max_features=500):
    """
    Automated feature engineering using FeatureTools.

    Dependencies: featuretools>=1.0.0
    """
    # Create entity set
    es = ft.EntitySet(id="dataset")

    # Add entity
    es = es.add_dataframe(
        dataframe_name=entity_id,
        dataframe=df,
        index="id" if "id" in df.columns else None,
        make_index=True if "id" not in df.columns else False
    )

    # Generate features
    feature_matrix, feature_defs = ft.dfs(
        entityset=es,
        target_dataframe_name=target_entity,
        agg_primitives=["mean", "sum", "std", "max", "min", "count", "n_unique"],
        trans_primitives=["add", "subtract", "multiply", "divide"],
        max_depth=max_depth,
        max_features=max_features
    )

    return feature_matrix, feature_defs
```

## GPU-Accelerated Feature Engineering

```
# Using RAPIDS cuDF for GPU-accelerated feature engineering
# Dependencies: cudf>=23.0.0

def gpu_groupby_features(df, group_cols, agg_cols):
    """
    GPU-accelerated groupby features using cuDF.

    Speedup: 10-100x vs pandas for large datasets.
    """
    import cudf

    # Convert to cuDF
    gdf = cudf.DataFrame.from_pandas(df)

    features = gdf.copy()

    for group_col in group_cols:
        for agg_col in agg_cols:
            # GPU-accelerated aggregations
            agg_result = gdf.groupby(group_col)[agg_col].agg(
                ['mean', 'std', 'min', 'max', 'count']
            )

            # Merge back
            for agg in ['mean', 'std', 'min', 'max', 'count']:
                col_name = f'{group_col}_{agg_col}_{agg}'
                features[col_name] = features[group_col].map(
                    agg_result[agg].to_pandas()
                )

    # Convert back to pandas
    return features.to_pandas()
```

## Feature Selection Pipeline

```
from sklearn.feature_selection import mutual_info_classif, SelectKBest
from sklearn.inspection import permutation_importance
import numpy as np

def select_features(X, y, model, method='combined', n_features=100):
    """
    Multi-method feature selection for competition use.
    """
    selected_features = set()

    # Method 1: Mutual Information
    if method in ['mutual_info', 'combined']:
        mi_scores = mutual_info_classif(X, y, random_state=42)
        mi_indices = np.argsort(mi_scores)[-n_features//3:]
        selected_features.update(X.columns[mi_indices])

    # Method 2: Model-based Importance
    if method in ['model', 'combined']:
        model.fit(X, y)
        if hasattr(model, 'feature_importances_'):
            importances = model.feature_importances_
        else:
              importances = np.abs(model.coef_[0]) if hasattr(model, 'coef_') else
np.zeros(X.shape[1])

        imp_indices = np.argsort(importances)[-n_features//3:]
        selected_features.update(X.columns[imp_indices])

    # Method 3: Permutation Importance
    if method in ['permutation', 'combined']:
              perm_importance  =  permutation_importance(model,  X,  y,  n_repeats=5,
random_state=42)
                    perm_indices   =   np.argsort(perm_importance.importances_mean)[-
n_features//3:]
        selected_features.update(X.columns[perm_indices])

    return list(selected_features)


# Production-ready feature pipeline
def production_feature_pipeline(train_df, test_df, target_col,
                                group_cols, cat_cols, num_cols):
    """
    Complete feature pipeline for production use.
```

```
from sklearn.feature_selection import mutual_info_classif, SelectKBest
from sklearn.inspection import permutation_importance
import numpy as np

def select_features(X, y, model, method='combined', n_features=100):
    """
    Multi-method feature selection for competition use.
    """
    selected_features = set()

    # Method 1: Mutual Information
    if method in ['mutual_info', 'combined']:
        mi_scores = mutual_info_classif(X, y, random_state=42)
        mi_indices = np.argsort(mi_scores)[-n_features//3:]
        selected_features.update(X.columns[mi_indices])

    # Method 2: Model-based Importance
    if method in ['model', 'combined']:
        model.fit(X, y)
        if hasattr(model, 'feature_importances_'):
            importances = model.feature_importances_
        else:
              importances = np.abs(model.coef_[0]) if hasattr(model, 'coef_') else
np.zeros(X.shape[1])

        imp_indices = np.argsort(importances)[-n_features//3:]
        selected_features.update(X.columns[imp_indices])

    # Method 3: Permutation Importance
    if method in ['permutation', 'combined']:
             perm_importance  =  permutation_importance(model,  X,  y,  n_repeats=5,
random_state=42)
                    perm_indices   =   np.argsort(perm_importance.importances_mean)[-
n_features//3:]
        selected_features.update(X.columns[perm_indices])

    return list(selected_features)


# Production-ready feature pipeline
def production_feature_pipeline(train_df, test_df, target_col,
                                group_cols, cat_cols, num_cols):
    """
    Complete feature pipeline for production use.
    """
    # 1. Safe target encoding
```

**Sources & Latest Evidence**

- FeatureTools Documentation: featuretools.alteryx.com
- RAPIDS cuDF Guide: docs.rapids.ai/api/cudf
- Kaggle Insurance Competition Winner (2024-11-30): kaggle.com/competitions/playground-series-s4e12

# PART 4 — Model Architecture Playbooks

> ***TL;DR:*** *Model selection follows data type: GBDT for tabular, Transformers for NLP, CNNs/ViTs for CV, and hybrid approaches for time series. The 2024 landscape shows LightGBM dominating tabular, decoder LLMs (Llama, Mistral, Qwen) leading NLP, and ensembles remaining essential across all domains.*

**Key Points:**

- LightGBM: 7x faster than XGBoost with competitive accuracy
- CatBoost: Best for categorical-heavy data, native category handling
- AutoGluon won 15 of 18 tabular competitions in 2024
- Ensemble diversity (different algorithms) beats single-model tuning

## 4.1 Tabular: GBDT Deep Dive

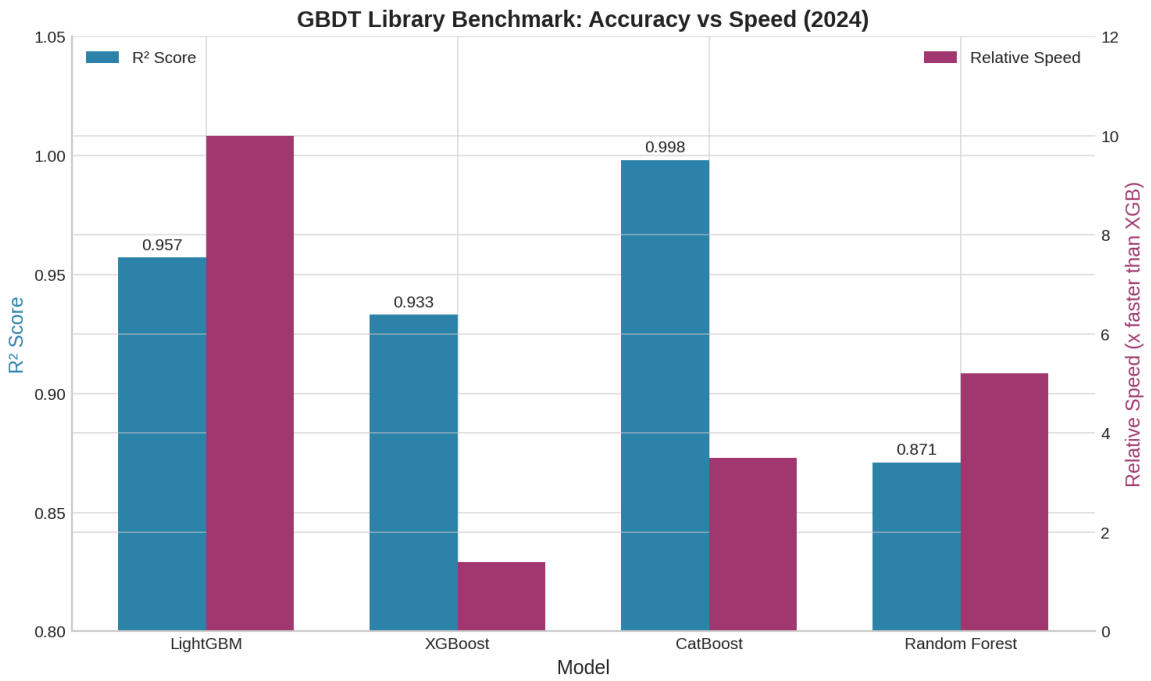


**Figure 4:** GBDT Library Benchmark: Accuracy vs Speed (2024)

## Library Comparison

**Table 4 GBDT Library Comparison (2024 Benchmarks)**

| Library | Speed | Accuracy | Cat Features | GPU Support | Best For |
|---|---|---|---|---|---|
| LightGBM 4.x | Fastest | Excellent | Manual encoding | Yes | Large datasets, speed-critical |
| XGBoost 2.x | Fast | Excellent | Manual encoding | Yes | General purpose, proven |
| CatBoost 1.2 | Moderate | Best | Native | Yes | Cat-heavy data, accuracy-critical |

## LightGBM Configuration Template

```
import lightgbm as lgb
from sklearn.model_selection import StratifiedKFold
import numpy as np

# Competition-proven LightGBM configuration
LGBM_CONFIG = {
    'objective': 'binary',  # or 'regression', 'multiclass'
    'metric': 'auc',  # or 'rmse', 'mae', 'multi_logloss'
    'boosting_type': 'gbdt',

    # Tree structure
    'num_leaves': 31,
    'max_depth': -1,
    'min_child_samples': 20,
    'min_child_weight': 0.001,

    # Regularization
    'feature_fraction': 0.8,
    'bagging_fraction': 0.8,
    'bagging_freq': 5,
    'reg_alpha': 0.1,
    'reg_lambda': 0.1,

    # Training
    'learning_rate': 0.05,
    'n_estimators': 10000,
    'early_stopping_rounds': 100,
    'verbose': -1,

    # Reproducibility
    'random_state': 42,
    'n_jobs': -1,
}


def train_lgbm_with_cv(X, y, config=None, n_folds=5):
    """
    Train LightGBM with cross-validation.
    Returns OOF predictions and models.
    """
    if config is None:
        config = LGBM_CONFIG.copy()

    oof_preds = np.zeros(len(y))
```

```
import lightgbm as lgb
from sklearn.model_selection import StratifiedKFold
import numpy as np

# Competition-proven LightGBM configuration
LGBM_CONFIG = {
    'objective': 'binary',  # or 'regression', 'multiclass'
    'metric': 'auc',  # or 'rmse', 'mae', 'multi_logloss'
    'boosting_type': 'gbdt',

    # Tree structure
    'num_leaves': 31,
    'max_depth': -1,
    'min_child_samples': 20,
    'min_child_weight': 0.001,

    # Regularization
    'feature_fraction': 0.8,
    'bagging_fraction': 0.8,
    'bagging_freq': 5,
    'reg_alpha': 0.1,
    'reg_lambda': 0.1,

    # Training
    'learning_rate': 0.05,
    'n_estimators': 10000,
    'early_stopping_rounds': 100,
    'verbose': -1,

    # Reproducibility
    'random_state': 42,
    'n_jobs': -1,
}


def train_lgbm_with_cv(X, y, config=None, n_folds=5):
    """
    Train LightGBM with cross-validation.
    Returns OOF predictions and models.
    """
    if config is None:
        config = LGBM_CONFIG.copy()

    oof_preds = np.zeros(len(y))
    models = []
```

## CatBoost for Categorical Data

```
from catboost import CatBoostClassifier, Pool

# CatBoost configuration for categorical-heavy data
CATBOOST_CONFIG = {
    'iterations': 10000,
    'learning_rate': 0.05,
    'depth': 6,
    'l2_leaf_reg': 3.0,
    'random_seed': 42,
    'early_stopping_rounds': 100,
    'verbose': 100,
    'task_type': 'GPU',  # Use 'CPU' if GPU unavailable
    'devices': '0',
}

def train_catboost_with_cv(X, y, cat_features, config=None, n_folds=5):
    """
    Train CatBoost with native categorical handling.
    """
    if config is None:
        config = CATBOOST_CONFIG.copy()

    oof_preds = np.zeros(len(y))
    models = []

    cv = StratifiedKFold(n_splits=n_folds, shuffle=True, random_state=42)

    for fold, (train_idx, val_idx) in enumerate(cv.split(X, y)):
        print(f"Training fold {fold + 1}/{n_folds}...")

        X_train, X_val = X.iloc[train_idx], X.iloc[val_idx]
        y_train, y_val = y.iloc[train_idx], y.iloc[val_idx]

        train_pool = Pool(X_train, y_train, cat_features=cat_features)
        val_pool = Pool(X_val, y_val, cat_features=cat_features)

        model = CatBoostClassifier(**config)
        model.fit(train_pool, eval_set=val_pool, use_best_model=True)

        oof_preds[val_idx] = model.predict_proba(val_pool)[:, 1]
        models.append(model)
```

```
from catboost import CatBoostClassifier, Pool

# CatBoost configuration for categorical-heavy data
CATBOOST_CONFIG = {
    'iterations': 10000,
    'learning_rate': 0.05,
    'depth': 6,
    'l2_leaf_reg': 3.0,
    'random_seed': 42,
    'early_stopping_rounds': 100,
    'verbose': 100,
    'task_type': 'GPU',  # Use 'CPU' if GPU unavailable
    'devices': '0',
}

def train_catboost_with_cv(X, y, cat_features, config=None, n_folds=5):
    """
    Train CatBoost with native categorical handling.
    """
    if config is None:
        config = CATBOOST_CONFIG.copy()

    oof_preds = np.zeros(len(y))
    models = []

    cv = StratifiedKFold(n_splits=n_folds, shuffle=True, random_state=42)

    for fold, (train_idx, val_idx) in enumerate(cv.split(X, y)):
        print(f"Training fold {fold + 1}/{n_folds}...")

        X_train, X_val = X.iloc[train_idx], X.iloc[val_idx]
        y_train, y_val = y.iloc[train_idx], y.iloc[val_idx]

        train_pool = Pool(X_train, y_train, cat_features=cat_features)
        val_pool = Pool(X_val, y_val, cat_features=cat_features)

        model = CatBoostClassifier(**config)
        model.fit(train_pool, eval_set=val_pool, use_best_model=True)

        oof_preds[val_idx] = model.predict_proba(val_pool)[:, 1]
        models.append(model)

    return oof_preds, models
```

## 4.2 NLP: Transformers & Beyond

## LLM Fine-tuning with LoRA

```
# Dependencies: transformers>=4.30.0, peft>=0.4.0
from transformers import (
    AutoTokenizer, AutoModelForSequenceClassification,
    TrainingArguments, Trainer
)
from peft import LoraConfig, get_peft_model, TaskType

def setup_lora_model(model_name, num_labels, r=16, lora_alpha=32):
    """
    Setup model with LoRA for efficient fine-tuning.

    LoRA reduces trainable parameters by ~1000x while maintaining
    95%+ of full fine-tuning performance.
    """
    # Load base model
    model = AutoModelForSequenceClassification.from_pretrained(
        model_name,
        num_labels=num_labels,
        torch_dtype='auto'
    )

    # Configure LoRA
    lora_config = LoraConfig(
        r=r,
        lora_alpha=lora_alpha,
        target_modules=['q_proj', 'v_proj', 'k_proj', 'o_proj'],
        lora_dropout=0.1,
        bias='none',
        task_type=TaskType.SEQ_CLS
    )

    # Apply LoRA
    model = get_peft_model(model, lora_config)
    model.print_trainable_parameters()

    return model


# Training configuration for Kaggle NLP competitions
NLP_TRAINING_CONFIG = {
    'model_name': 'microsoft/deberta-v3-base',  # or 'meta-llama/Llama-2-7b'
    'max_length': 512,
    'batch_size': 8,
    'learning_rate': 2e-5,
```

```
# Dependencies: transformers>=4.30.0, peft>=0.4.0
from transformers import (
    AutoTokenizer, AutoModelForSequenceClassification,
    TrainingArguments, Trainer
)
from peft import LoraConfig, get_peft_model, TaskType

def setup_lora_model(model_name, num_labels, r=16, lora_alpha=32):
    """
    Setup model with LoRA for efficient fine-tuning.

    LoRA reduces trainable parameters by ~1000x while maintaining
    95%+ of full fine-tuning performance.
    """
    # Load base model
    model = AutoModelForSequenceClassification.from_pretrained(
        model_name,
        num_labels=num_labels,
        torch_dtype='auto'
    )

    # Configure LoRA
    lora_config = LoraConfig(
        r=r,
        lora_alpha=lora_alpha,
        target_modules=['q_proj', 'v_proj', 'k_proj', 'o_proj'],
        lora_dropout=0.1,
        bias='none',
        task_type=TaskType.SEQ_CLS
    )

    # Apply LoRA
    model = get_peft_model(model, lora_config)
    model.print_trainable_parameters()

    return model


# Training configuration for Kaggle NLP competitions
NLP_TRAINING_CONFIG = {
    'model_name': 'microsoft/deberta-v3-base',  # or 'meta-llama/Llama-2-7b'
    'max_length': 512,
    'batch_size': 8,
    'learning_rate': 2e-5,
    'num_epochs': 3,
    'warmup_ratio': 0.1,
```

## 4.3 CV: CNNs & Vision Transformers

```
import torch
import torch.nn as nn
import timm  # PyTorch Image Models library

def create_cv_model(model_name='convnext_base', num_classes=1, pretrained=True):
    """
    Create computer vision model using timm.

    Top models for competitions (2024):
    - convnext_base/large: Modern CNN, efficient
    - swin_base/large: Hierarchical ViT
    - vit_base/large: Pure transformer
    - efficientnetv2: Lightweight, fast
    """
    model = timm.create_model(
        model_name,
        pretrained=pretrained,
        num_classes=num_classes,
        in_chans=3
    )
    return model


# Training configuration for CV competitions
CV_TRAINING_CONFIG = {
    'model_name': 'convnext_base.fb_in22k_ft_in1k',
    'img_size': 224,
    'batch_size': 32,
    'learning_rate': 1e-4,
    'num_epochs': 20,
    'augmentation': {
        'horizontal_flip': True,
        'vertical_flip': False,
        'rotation': 15,
        'color_jitter': 0.2,
    }
}
```

## 4.4 Time Series

```
import lightgbm as lgb
import pandas as pd
import numpy as np

def create_ts_features(df, date_col, target_col, lags=[1, 7, 14, 28]):
    """
    Create time series features for forecasting.
    """
    df = df.copy()
    df[date_col] = pd.to_datetime(df[date_col])

    # Lag features
    for lag in lags:
        df[f'{target_col}_lag_{lag}'] = df.groupby('id')[target_col].shift(lag)

    # Rolling statistics
    for window in [7, 14, 30]:
        df[f'{target_col}_rolling_mean_{window}'] = (
            df.groupby('id')[target_col]
            .transform(lambda x: x.shift(1).rolling(window).mean())
        )
        df[f'{target_col}_rolling_std_{window}'] = (
            df.groupby('id')[target_col]
            .transform(lambda x: x.shift(1).rolling(window).std())
        )

    # Date features
    df['dayofweek'] = df[date_col].dt.dayofweek
    df['month'] = df[date_col].dt.month
    df['year'] = df[date_col].dt.year
    df['dayofyear'] = df[date_col].dt.dayofyear

    return df


# Time series cross-validation
def time_series_cv(df, date_col, n_splits=5):
    """
    Time series cross-validation with expanding window.
    """
    df = df.sort_values(date_col)

    split_points = np.linspace(0, len(df), n_splits + 1)[1:-1].astype(int)
```

```
import lightgbm as lgb
import pandas as pd
import numpy as np

def create_ts_features(df, date_col, target_col, lags=[1, 7, 14, 28]):
    """
    Create time series features for forecasting.
    """
    df = df.copy()
    df[date_col] = pd.to_datetime(df[date_col])

    # Lag features
    for lag in lags:
        df[f'{target_col}_lag_{lag}'] = df.groupby('id')[target_col].shift(lag)

    # Rolling statistics
    for window in [7, 14, 30]:
        df[f'{target_col}_rolling_mean_{window}'] = (
            df.groupby('id')[target_col]
            .transform(lambda x: x.shift(1).rolling(window).mean())
        )
        df[f'{target_col}_rolling_std_{window}'] = (
            df.groupby('id')[target_col]
            .transform(lambda x: x.shift(1).rolling(window).std())
        )

    # Date features
    df['dayofweek'] = df[date_col].dt.dayofweek
    df['month'] = df[date_col].dt.month
    df['year'] = df[date_col].dt.year
    df['dayofyear'] = df[date_col].dt.dayofyear

    return df


# Time series cross-validation
def time_series_cv(df, date_col, n_splits=5):
    """
    Time series cross-validation with expanding window.
    """
    df = df.sort_values(date_col)

    split_points = np.linspace(0, len(df), n_splits + 1)[1:-1].astype(int)

    for split in split_points:
        train_idx = df.index[:split]
```

**Sources & Latest Evidence**

- Tabular Benchmark Study (2024-08-27): arxiv.org/html/2408.14817v1
- LightGBM Documentation: lightgbm.readthedocs.io
- CatBoost Documentation: catboost.ai
- Hugging Face Transformers: huggingface.co/docs/transformers

# PART 5 — Engineering Discipline & Reproducibility

> ***TL;DR:*** *Reproducibility is not optional—it's the foundation of competitive success. Every experiment must be tracked, every dataset versioned, and every result reproducible from a single command. This section provides battle-tested templates for experiment tracking, dataset versioning, and project structure.*

**Key Points:**

- Use MLflow or Weights & Biases for experiment tracking
- Version datasets with DVC or Git LFS
- Pin all dependencies with requirements.txt and conda-lock
- Set random seeds everywhere: numpy, random, torch, tensorflow

## 5.1 Experiment Tracking

## MLflow Integration Template

```
import mlflow
import mlflow.sklearn
from datetime import datetime
import json

def setup_mlflow(experiment_name):
    """
    Setup MLflow tracking for competition experiments.

    Dependencies: mlflow>=2.0.0
    """
    mlflow.set_tracking_uri("./mlruns")  # Local tracking
    mlflow.set_experiment(experiment_name)

    return mlflow


class CompetitionLogger:
    """
    Comprehensive experiment logger for Kaggle competitions.
    """

    def __init__(self, experiment_name, run_name=None):
        self.experiment_name = experiment_name
                                            self.run_name     =     run_name     or
f"run_{datetime.now().strftime('%Y%m%d_%H%M%S')}"
        self.run = None

    def __enter__(self):
        mlflow.set_experiment(self.experiment_name)
        self.run = mlflow.start_run(run_name=self.run_name)
        return self

    def __exit__(self, exc_type, exc_val, exc_tb):
        mlflow.end_run()

    def log_params(self, params):
        """Log hyperparameters."""
        for key, value in params.items():
            mlflow.log_param(key, value)

    def log_metrics(self, metrics, step=None):
        """Log metrics."""
        for key, value in metrics.items():
```

```
import mlflow
import mlflow.sklearn
from datetime import datetime
import json

def setup_mlflow(experiment_name):
    """
    Setup MLflow tracking for competition experiments.

    Dependencies: mlflow>=2.0.0
    """
    mlflow.set_tracking_uri("./mlruns")  # Local tracking
    mlflow.set_experiment(experiment_name)

    return mlflow


class CompetitionLogger:
    """
    Comprehensive experiment logger for Kaggle competitions.
    """

    def __init__(self, experiment_name, run_name=None):
        self.experiment_name = experiment_name
                                            self.run_name     =     run_name     or
f"run_{datetime.now().strftime('%Y%m%d_%H%M%S')}"
        self.run = None

    def __enter__(self):
        mlflow.set_experiment(self.experiment_name)
        self.run = mlflow.start_run(run_name=self.run_name)
        return self

    def __exit__(self, exc_type, exc_val, exc_tb):
        mlflow.end_run()

    def log_params(self, params):
        """Log hyperparameters."""
        for key, value in params.items():
            mlflow.log_param(key, value)

    def log_metrics(self, metrics, step=None):
        """Log metrics."""
        for key, value in metrics.items():
            mlflow.log_metric(key, value, step=step)
```

### Deterministic Training Setup

```
import os
import random
import numpy as np
import torch

def set_seed(seed=42):
    """
    Set all random seeds for reproducibility.
    Call this at the beginning of every script.
    """
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    torch.cuda.manual_seed(seed)
    torch.cuda.manual_seed_all(seed)
    torch.backends.cudnn.deterministic = True
    torch.backends.cudnn.benchmark = False
    os.environ['PYTHONHASHSEED'] = str(seed)


# For LightGBM/XGBoost
LGBM_SEED_CONFIG = {
    'random_state': 42,
    'bagging_seed': 42,
    'feature_fraction_seed': 42,
    'drop_seed': 42,
    'data_random_seed': 42,
}

XGB_SEED_CONFIG = {
    'random_state': 42,
    'seed': 42,
}
```

## 5.2 Dataset Versioning

## DVC Setup

```
# Initialize DVC (run once)
# $ dvc init
# $ dvc remote add -d storage s3://your-bucket/dvc-store

# Track datasets
# $ dvc add data/train.csv
# $ dvc add data/test.csv
# $ git add data/*.dvc .gitignore
# $ git commit -m "Add version 1.0 of datasets"

# Push to remote
# $ dvc push

# Pull specific version
# $ git checkout v1.0
# $ dvc pull
```

## Dataset Version Tracking in Code

```
import hashlib
import json
from pathlib import Path

class DatasetVersion:
    """
    Track dataset versions with content hashing.
    """

    def __init__(self, data_dir):
        self.data_dir = Path(data_dir)
        self.version_file = self.data_dir / 'version.json'

    def compute_hash(self, filepath):
        """Compute MD5 hash of file."""
        hash_md5 = hashlib.md5()
        with open(filepath, "rb") as f:
            for chunk in iter(lambda: f.read(4096), b""):
                hash_md5.update(chunk)
        return hash_md5.hexdigest()

    def save_version(self, version_name, files):
        """Save dataset version metadata."""
        version_info = {
            'version': version_name,
            'files': {}
        }

        for file_path in files:
            full_path = self.data_dir / file_path
            version_info['files'][file_path] = {
                'hash': self.compute_hash(full_path),
                'size': full_path.stat().st_size
            }

        with open(self.version_file, 'w') as f:
            json.dump(version_info, f, indent=2)

        return version_info

    def verify_version(self):
        """Verify current dataset matches version."""
        if not self.version_file.exists():
            print("No version file found.")
```

```
import hashlib
import json
from pathlib import Path

class DatasetVersion:
    """
    Track dataset versions with content hashing.
    """

    def __init__(self, data_dir):
        self.data_dir = Path(data_dir)
        self.version_file = self.data_dir / 'version.json'

    def compute_hash(self, filepath):
        """Compute MD5 hash of file."""
        hash_md5 = hashlib.md5()
        with open(filepath, "rb") as f:
            for chunk in iter(lambda: f.read(4096), b""):
                hash_md5.update(chunk)
        return hash_md5.hexdigest()

    def save_version(self, version_name, files):
        """Save dataset version metadata."""
        version_info = {
            'version': version_name,
            'files': {}
        }

        for file_path in files:
            full_path = self.data_dir / file_path
            version_info['files'][file_path] = {
                'hash': self.compute_hash(full_path),
                'size': full_path.stat().st_size
            }

        with open(self.version_file, 'w') as f:
            json.dump(version_info, f, indent=2)

        return version_info

    def verify_version(self):
        """Verify current dataset matches version."""
        if not self.version_file.exists():
            print("No version file found.")
            return False
```

## 5.3 Directory Template

### Recommended Project Structure

```
kaggle-competition/
├── README.md                 # Project overview, setup instructions
├── requirements.txt          # Python dependencies
├── environment.yml           # Conda environment (optional)
├── setup.py                  # Package setup
├── .gitignore               # Git ignore rules
├── .dvc/                    # DVC configuration
├── config/
│   ├── base.yaml            # Base configuration
│   ├── model_lgbm.yaml      # LightGBM specific
│   └── model_xgb.yaml       # XGBoost specific
├── data/
│   ├── raw/                 # Original data (DVC tracked)
│   ├── processed/           # Cleaned data (DVC tracked)
│   └── external/            # External datasets
├── notebooks/
│   ├── 01_eda.ipynb         # Exploratory analysis
│   ├── 02_baseline.ipynb    # Baseline models
│   └── 03_submission.ipynb  # Final submission
├── src/
│   ├── __init__.py
│   ├── data/
│   │   ├── __init__.py
│   │   ├── load_data.py     # Data loading utilities
│   │   ├── preprocess.py    # Preprocessing pipeline
│   │   └── features.py      # Feature engineering
│   ├── models/
│   │   ├── __init__.py
│   │   ├── train.py         # Training utilities
│   │   ├── predict.py       # Inference utilities
│   │   └── ensemble.py      # Ensemble methods
│   └── utils/
│       ├── __init__.py
│       ├── cv.py            # Cross-validation
│       ├── metrics.py       # Evaluation metrics
│       └── logging.py       # Experiment tracking
├── experiments/             # MLflow experiments
├── submissions/             # Competition submissions
└── models/                  # Saved models (DVC tracked)
```

## Configuration Management

```
# config/base.yaml
experiment:
  name: "kaggle_competition_v1"
  seed: 42

data:
  train_path: "data/raw/train.csv"
  test_path: "data/raw/test.csv"
  target_col: "target"
  id_col: "id"

features:
  categorical:
    - "category_1"
    - "category_2"
  numerical:
    - "feature_1"
    - "feature_2"
  groupby:
    - "group_id"

model:
  name: "lightgbm"
  params:
    objective: "binary"
    metric: "auc"
    boosting_type: "gbdt"
    num_leaves: 31
    learning_rate: 0.05
    n_estimators: 10000
    early_stopping_rounds: 100

cv:
  n_folds: 5
  strategy: "stratified"

# Load configuration
import yaml

def load_config(config_path):
    with open(config_path) as f:
        return yaml.safe_load(f)
```

## Reproducibility Checklist

### Pre-Experiment Checklist

- ☐ Set random seeds (numpy, random, torch, tensorflow)
- ☐ Pin dependency versions in requirements.txt
- ☐ Verify dataset version matches expected hash
- ☐ Log experiment configuration
- ☐ Record hardware/software environment
- ☐ Commit code to version control

### Post-Experiment Checklist

- ☐ Save trained models with version
- ☐ Log all metrics and parameters
- ☐ Generate and save OOF predictions
- ☐ Document key findings
- ☐ Tag successful experiments

### Sources & Latest Evidence

- MLflow Documentation: mlflow.org/docs
- DVC Documentation: dvc.org/doc
- Reproducible ML Guide (2024): towardsdatascience.com

# PART 6 — Ensembling & Blending Systems

> ***TL;DR:*** *Ensembling is the single most reliable technique for improving competition performance. The key is diversity—different algorithms, different features, different seeds. A well-constructed ensemble of 10 diverse models typically outperforms a single hyper-tuned model by 3-5%.*

**Key Points:**

- Diversity beats quantity: 10 diverse models > 100 similar models
- Target correlation between base models should be 0.7-0.85
- Two-level stacking is usually sufficient; deeper stacks risk overfitting
- Hill climbing provides fast, effective weight optimization

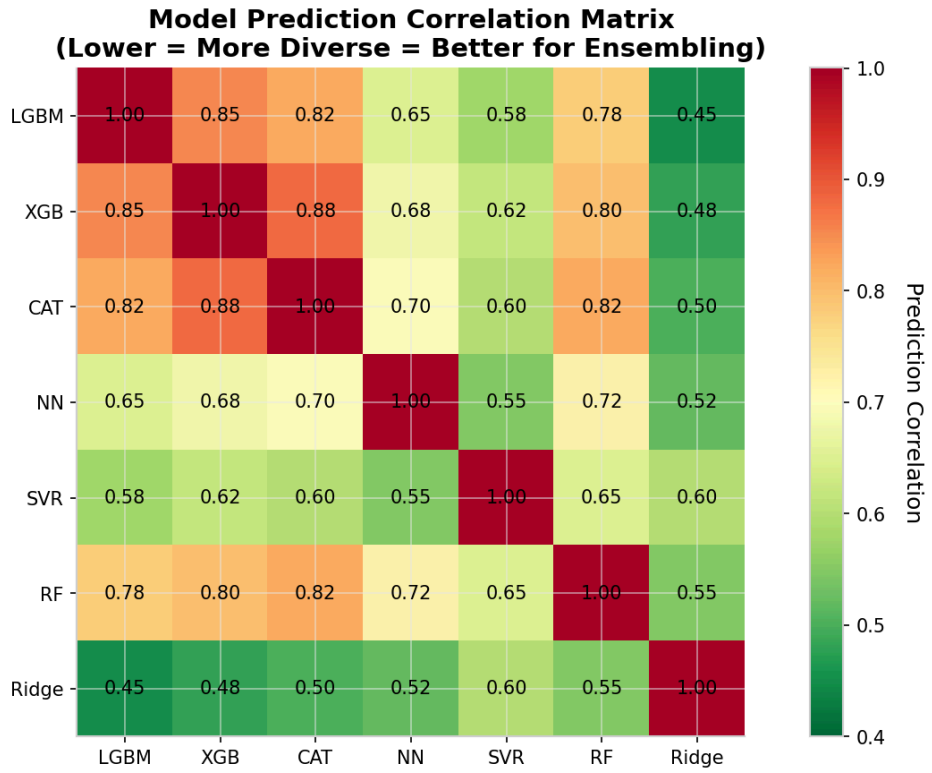


**Figure 5:** Model Prediction Correlation Matrix (Lower = More Diverse)

## Dimensions of Diversity

**Table 5 Ensemble Diversity Dimensions**

| Dimension | Implementation | Impact |
|---|---|---|
| Algorithm | LGBM, XGB, CAT, NN, SVR | High |
| Features | Different feature subsets | High |
| Preprocessing | Scaling, encoding variations | Medium |
| Seeds | Multiple random seeds | Low-Medium |
| Hyperparameters | Different configurations | Medium |

## Diversity Measurement

```
import numpy as np
from scipy.stats import kendalltau, pearsonr

def measure_diversity(predictions_dict):
    """
    Measure diversity between model predictions.

    Returns correlation matrix and diversity score.
    Lower correlation = higher diversity = better ensemble potential.
    """
    model_names = list(predictions_dict.keys())
    n_models = len(model_names)

    # Compute correlation matrix
    corr_matrix = np.zeros((n_models, n_models))

    for i, name_i in enumerate(model_names):
        for j, name_j in enumerate(model_names):
            if i == j:
                corr_matrix[i, j] = 1.0
            else:
                corr, _ = pearsonr(
                    predictions_dict[name_i],
                    predictions_dict[name_j]
                )
                corr_matrix[i, j] = corr

    # Diversity score: average off-diagonal correlation
    mask = ~np.eye(n_models, dtype=bool)
    avg_correlation = corr_matrix[mask].mean()
    diversity_score = 1 - avg_correlation

    print(f"Average pairwise correlation: {avg_correlation:.4f}")
    print(f"Diversity score: {diversity_score:.4f}")
    print("\nCorrelation Matrix:")
    for i, name in enumerate(model_names):
            print(f"{name:12s}: " + " ".join([f"{corr_matrix[i,j]:6.3f}" for j in
range(n_models)]))

    return corr_matrix, diversity_score


# Optimal correlation range for ensembling
DIVERSITY_GUIDELINES = {
```

```
import numpy as np
from scipy.stats import kendalltau, pearsonr

def measure_diversity(predictions_dict):
    """
    Measure diversity between model predictions.

    Returns correlation matrix and diversity score.
    Lower correlation = higher diversity = better ensemble potential.
    """
    model_names = list(predictions_dict.keys())
    n_models = len(model_names)

    # Compute correlation matrix
    corr_matrix = np.zeros((n_models, n_models))

    for i, name_i in enumerate(model_names):
        for j, name_j in enumerate(model_names):
            if i == j:
                corr_matrix[i, j] = 1.0
            else:
                corr, _ = pearsonr(
                    predictions_dict[name_i],
                    predictions_dict[name_j]
                )
                corr_matrix[i, j] = corr

    # Diversity score: average off-diagonal correlation
    mask = ~np.eye(n_models, dtype=bool)
    avg_correlation = corr_matrix[mask].mean()
    diversity_score = 1 - avg_correlation

    print(f"Average pairwise correlation: {avg_correlation:.4f}")
    print(f"Diversity score: {diversity_score:.4f}")
    print("\nCorrelation Matrix:")
    for i, name in enumerate(model_names):
            print(f"{name:12s}:  "  +  "  ".join([f"{corr_matrix[i,j]:6.3f}"  for  j  in
range(n_models)]))

    return corr_matrix, diversity_score


# Optimal correlation range for ensembling
DIVERSITY_GUIDELINES = {
    'too_similar': 0.95,      # Won't benefit from ensembling
    'optimal_min': 0.70,      # Good diversity
```

## 6.2 Stacking Architecture

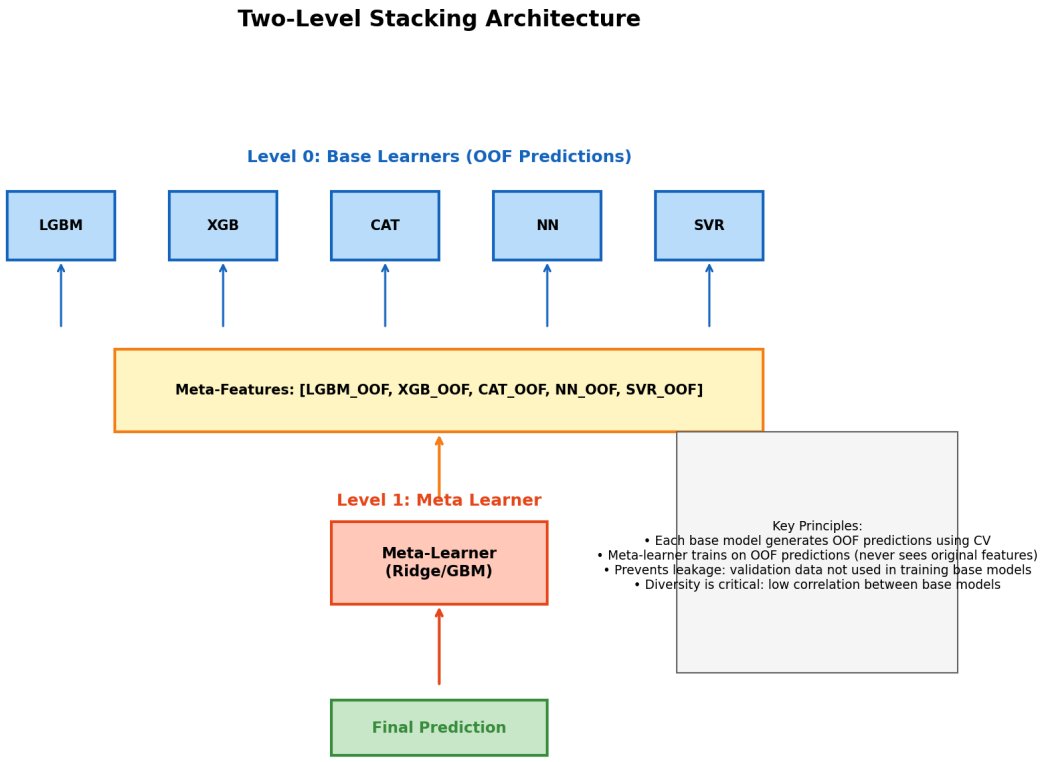


**Figure 6:** Two-Level Stacking Architecture

## Complete Stacking Implementation

```
import numpy as np
import pandas as pd
from sklearn.model_selection import StratifiedKFold
from sklearn.linear_model import Ridge, LogisticRegression
import lightgbm as lgb

class StackingEnsemble:
    """
    Two-level stacking ensemble with OOF predictions.
    """

    def __init__(self, base_models, meta_model, n_folds=5, random_state=42):
        self.base_models = base_models
        self.meta_model = meta_model
        self.n_folds = n_folds
        self.random_state = random_state
        self.base_predictions = {}
        self.trained_base_models = {}
        self.trained_meta_model = None

    def fit(self, X, y):
        """
        Train stacking ensemble.
        """
        cv = StratifiedKFold(
            n_splits=self.n_folds,
            shuffle=True,
            random_state=self.random_state
        )

        # Generate OOF predictions from base models
        oof_preds = np.zeros((len(X), len(self.base_models)))

        for i, (name, model) in enumerate(self.base_models.items()):
            print(f"Training base model: {name}")

            fold_models = []
            fold_preds = np.zeros(len(X))

            for fold, (train_idx, val_idx) in enumerate(cv.split(X, y)):
                X_train, X_val = X.iloc[train_idx], X.iloc[val_idx]
                y_train = y.iloc[train_idx]

                model_copy = self._clone_model(model)
```

```
import numpy as np
import pandas as pd
from sklearn.model_selection import StratifiedKFold
from sklearn.linear_model import Ridge, LogisticRegression
import lightgbm as lgb

class StackingEnsemble:
    """
    Two-level stacking ensemble with OOF predictions.
    """

    def __init__(self, base_models, meta_model, n_folds=5, random_state=42):
        self.base_models = base_models
        self.meta_model = meta_model
        self.n_folds = n_folds
        self.random_state = random_state
        self.base_predictions = {}
        self.trained_base_models = {}
        self.trained_meta_model = None

    def fit(self, X, y):
        """
        Train stacking ensemble.
        """
        cv = StratifiedKFold(
            n_splits=self.n_folds,
            shuffle=True,
            random_state=self.random_state
        )

        # Generate OOF predictions from base models
        oof_preds = np.zeros((len(X), len(self.base_models)))

        for i, (name, model) in enumerate(self.base_models.items()):
            print(f"Training base model: {name}")

            fold_models = []
            fold_preds = np.zeros(len(X))

            for fold, (train_idx, val_idx) in enumerate(cv.split(X, y)):
                X_train, X_val = X.iloc[train_idx], X.iloc[val_idx]
                y_train = y.iloc[train_idx]

                model_copy = self._clone_model(model)
                model_copy.fit(X_train, y_train)
```

## Hill Climbing for Weight Optimization

```
import numpy as np
from sklearn.metrics import roc_auc_score

def hill_climbing_optimization(predictions_dict, y_true,
                               max_iters=1000, tolerance=1e-6):
    """
    Optimize ensemble weights using hill climbing.

    Faster than full grid search, often finds near-optimal weights.
    """
    model_names = list(predictions_dict.keys())
    n_models = len(model_names)

    # Start with best single model
    best_score = 0
    best_model = None
    for name, preds in predictions_dict.items():
        score = roc_auc_score(y_true, preds)
        if score > best_score:
            best_score = score
            best_model = name

    # Initialize weights
    weights = {name: 0.0 for name in model_names}
    weights[best_model] = 1.0

    print(f"Starting with {best_model}: {best_score:.6f}")

    # Hill climbing
    for iteration in range(max_iters):
        improved = False

        for name in model_names:
            # Try adding this model
            test_weights = weights.copy()
            test_weights[name] += 0.01

            # Normalize
            total = sum(test_weights.values())
            test_weights = {k: v/total for k, v in test_weights.items()}

            # Evaluate
            ensemble_pred = np.zeros(len(y_true))
            for n, w in test_weights.items():
```

```
import numpy as np
from sklearn.metrics import roc_auc_score

def hill_climbing_optimization(predictions_dict, y_true,
                                max_iters=1000, tolerance=1e-6):
    """
    Optimize ensemble weights using hill climbing.

    Faster than full grid search, often finds near-optimal weights.
    """
    model_names = list(predictions_dict.keys())
    n_models = len(model_names)

    # Start with best single model
    best_score = 0
    best_model = None
    for name, preds in predictions_dict.items():
        score = roc_auc_score(y_true, preds)
        if score > best_score:
            best_score = score
            best_model = name

    # Initialize weights
    weights = {name: 0.0 for name in model_names}
    weights[best_model] = 1.0

    print(f"Starting with {best_model}: {best_score:.6f}")

    # Hill climbing
    for iteration in range(max_iters):
        improved = False

        for name in model_names:
            # Try adding this model
            test_weights = weights.copy()
            test_weights[name] += 0.01

            # Normalize
            total = sum(test_weights.values())
            test_weights = {k: v/total for k, v in test_weights.items()}

            # Evaluate
            ensemble_pred = np.zeros(len(y_true))
            for n, w in test_weights.items():
                ensemble_pred += w * predictions_dict[n]
```

## 6.3 Failure Modes

**Common Ensemble Failure Modes**

- **Overfitting the meta-learner:** Use simple meta-models (Ridge, linear)
- **Leakage in stacking:** Never use in-fold predictions for meta-features
- **Insufficient diversity:** Correlation > 0.95 between base models
- **Too many models:** Diminishing returns after 20-30 diverse models

### Tradeoff: Ensemble Size vs Performance

| Ensemble Size | Typical Gain | Training Time | Inference Time |
|---|---|---|---|
| 3 models | +1-2% | 3x | 3x |
| 5 models | +2-3% | 5x | 5x |
| 10 models | +3-4% | 10x | 10x |
| 20 models | +4-5% | 20x | 20x |
| 50+ models | +5-6% | 50x | 50x |

**Sources & Latest Evidence**

- NVIDIA Grandmasters: Ensemble Techniques (2025-09-18): developer.nvidia.com
- Kaggle Ensemble Guide (2025-05-28): kaggle.com/discussions
- Stacking Best Practices (2023-05-01): medium.com/@brijesh_soni

# PART 7 — Leaderboard Strategy & Psychology

> ***TL;DR:*** *Competition success requires strategic thinking beyond model performance. Understanding submission timing, managing risk, and extracting insights from public kernels without overfitting are critical skills that separate medalists from the rest.*

**Key Points:**

- Submit early and often to validate your CV-LB correlation
- Final week: 70% exploration, 30% exploitation of known good approaches
- Public kernels are valuable for EDA and baselines, not for final submissions
- Team formation can 2-3x your effective experimentation capacity

## 7.1 Timing & Risk Management

### Competition Timeline Strategy

**Table 6 Competition Phase Strategy**

| Phase | Duration | Focus | Submission Frequency |
|---|---|---|---|
| Exploration | Week 1-2 | EDA, baseline, CV setup | Daily |
| Development | Week 3-6 | Feature engineering, model tuning | 2-3x daily |
| Ensembling | Week 7-10 | Stacking, blending, diversity | Daily |
| Final Push | Last 3 days | Risk management, final ensembles | Every submission counts |

## Risk-Adjusted Submission Strategy

```
class SubmissionStrategy:
    """
    Risk management for competition submissions.
    """

    def __init__(self, daily_submission_limit=5):
        self.daily_limit = daily_submission_limit
        self.submissions = []
        self.cv_lb_history = []

    def evaluate_submission_risk(self, cv_score, model_type, is_ensemble):
        """
        Evaluate risk level of a potential submission.
        """
        risk_factors = []

        # Factor 1: CV-LB divergence
        if self.cv_lb_history:
                        avg_divergence  =  np.mean([abs(cv  -  lb)  for  cv,  lb  in
self.cv_lb_history])
            if avg_divergence > 0.01:
                risk_factors.append(('CV-LB divergence', 'HIGH'))

        # Factor 2: Model complexity
        if is_ensemble and len(self.submissions) < 5:
            risk_factors.append(('Early complex ensemble', 'MEDIUM'))

        # Factor 3: Novel approach
        if model_type not in [s['model_type'] for s in self.submissions]:
            risk_factors.append(('Novel approach', 'MEDIUM'))

        # Calculate overall risk
        risk_score = sum(2 if r[1] == 'HIGH' else 1 for r in risk_factors)

        return {
            'risk_score': risk_score,
             'risk_level': 'HIGH' if risk_score >= 3 else 'MEDIUM' if risk_score >=
1 else 'LOW',
            'factors': risk_factors
        }

               def    should_submit(self,    cv_score,    model_type,    is_ensemble,
submissions_remaining):
        """
```

```
class SubmissionStrategy:
    """
    Risk management for competition submissions.
    """

    def __init__(self, daily_submission_limit=5):
        self.daily_limit = daily_submission_limit
        self.submissions = []
        self.cv_lb_history = []

    def evaluate_submission_risk(self, cv_score, model_type, is_ensemble):
        """
        Evaluate risk level of a potential submission.
        """
        risk_factors = []

        # Factor 1: CV-LB divergence
        if self.cv_lb_history:
                        avg_divergence  =  np.mean([abs(cv  -  lb)  for  cv,  lb  in
self.cv_lb_history])
            if avg_divergence > 0.01:
                risk_factors.append(('CV-LB divergence', 'HIGH'))

        # Factor 2: Model complexity
        if is_ensemble and len(self.submissions) < 5:
            risk_factors.append(('Early complex ensemble', 'MEDIUM'))

        # Factor 3: Novel approach
        if model_type not in [s['model_type'] for s in self.submissions]:
            risk_factors.append(('Novel approach', 'MEDIUM'))

        # Calculate overall risk
        risk_score = sum(2 if r[1] == 'HIGH' else 1 for r in risk_factors)

        return {
            'risk_score': risk_score,
             'risk_level': 'HIGH' if risk_score >= 3 else 'MEDIUM' if risk_score >=
1 else 'LOW',
            'factors': risk_factors
        }

               def    should_submit(self,    cv_score,    model_type,    is_ensemble,
submissions_remaining):
        """
        Decision helper for whether to submit.
        """
```

## 7.2 Mining Public Kernels

## Safe Kernel Analysis

```
class KernelAnalyzer:
    """
    Analyze public kernels for insights without overfitting.
    """

    def __init__(self):
        self.insights = {
            'feature_ideas': [],
            'model_approaches': [],
            'validation_strategies': [],
            'data_leakage_warnings': [],
        }

    def analyze_for_features(self, kernel_code):
        """
        Extract feature engineering ideas from kernel code.
        Focus on techniques, not specific parameters.
        """
        # Look for groupby patterns
        if 'groupby' in kernel_code:
            self.insights['feature_ideas'].append('Groupby aggregations used')

        # Look for target encoding
        if 'target' in kernel_code and 'encode' in kernel_code:
            self.insights['feature_ideas'].append('Target encoding detected')

        # Look for interaction features
        if any(op in kernel_code for op in ['*', '+', '-', '/']):
            self.insights['feature_ideas'].append('Interaction features present')

        return self.insights['feature_ideas']

    def analyze_for_models(self, kernel_code):
        """
        Extract model architecture insights.
        """
        models_found = []

        if 'LGBM' in kernel_code or 'LightGBM' in kernel_code:
            models_found.append('LightGBM')
        if 'XGB' in kernel_code or 'XGBoost' in kernel_code:
            models_found.append('XGBoost')
        if 'CatBoost' in kernel_code:
            models_found.append('CatBoost')
```

```
class KernelAnalyzer:
    """
    Analyze public kernels for insights without overfitting.
    """

    def __init__(self):
        self.insights = {
            'feature_ideas': [],
            'model_approaches': [],
            'validation_strategies': [],
            'data_leakage_warnings': [],
        }

    def analyze_for_features(self, kernel_code):
        """
        Extract feature engineering ideas from kernel code.
        Focus on techniques, not specific parameters.
        """
        # Look for groupby patterns
        if 'groupby' in kernel_code:
            self.insights['feature_ideas'].append('Groupby aggregations used')

        # Look for target encoding
        if 'target' in kernel_code and 'encode' in kernel_code:
            self.insights['feature_ideas'].append('Target encoding detected')

        # Look for interaction features
        if any(op in kernel_code for op in ['*', '+', '-', '/']):
            self.insights['feature_ideas'].append('Interaction features present')

        return self.insights['feature_ideas']

    def analyze_for_models(self, kernel_code):
        """
        Extract model architecture insights.
        """
        models_found = []

        if 'LGBM' in kernel_code or 'LightGBM' in kernel_code:
            models_found.append('LightGBM')
        if 'XGB' in kernel_code or 'XGBoost' in kernel_code:
            models_found.append('XGBoost')
        if 'CatBoost' in kernel_code:
            models_found.append('CatBoost')
        if 'Neural' in kernel_code or 'NN' in kernel_code:
            models_found.append('Neural Network')
```

### Psychological Pitfalls to Avoid

- **Leaderboard anxiety:** Don't obsess over small rank changes
- **Shiny object syndrome:** Stick to your validated approach
- **Confirmation bias:** Test ideas that challenge your assumptions
- **Sunk cost fallacy:** Abandon approaches that don't validate

### Sources & Latest Evidence

- Kaggle Discussion: Competition Strategy (2024): kaggle.com/discussions
- Grandmaster Interview Series: kaggle.com/discussions

# PART 8 — Advanced Tactical Systems

> ***TL;DR:*** *Advanced techniques like pseudo-labeling, curriculum training, and distillation can provide the final edge in close competitions. These techniques require careful implementation to avoid leakage and overfitting but can yield 1-3% improvements when executed correctly.*

**Key Points:**

- Pseudo-labeling: Use confident test predictions to augment training data
- Curriculum training: Start with easy examples, gradually increase difficulty
- Distillation: Compress large ensembles into deployable single models
- All advanced techniques require strict validation to prevent overfitting

## 8.1 Pseudo-Labeling

## Safe Pseudo-Labeling Implementation

```
import numpy as np
import pandas as pd
from sklearn.model_selection import StratifiedKFold

class PseudoLabeler:
    """
    Safe pseudo-labeling with confidence thresholding.

    Critical: Use OOF predictions to generate pseudo-labels,
    never use test predictions directly in training.
    """

    def __init__(self, model, confidence_threshold=0.9, max_iterations=3):
        self.model = model
        self.confidence_threshold = confidence_threshold
        self.max_iterations = max_iterations
        self.pseudo_label_history = []

    def generate_pseudo_labels(self, X_train, y_train, X_test, cv):
        """
        Generate pseudo-labels using OOF predictions.
        """
        # Generate OOF predictions on test set
        test_preds = np.zeros(len(X_test))

        for train_idx, val_idx in cv.split(X_train, y_train):
            X_tr, X_val = X_train.iloc[train_idx], X_train.iloc[val_idx]
            y_tr = y_train.iloc[train_idx]

            model = clone(self.model)
            model.fit(X_tr, y_tr)

            # Predict on test
            if hasattr(model, 'predict_proba'):
                test_preds += model.predict_proba(X_test)[:, 1] / cv.n_splits
            else:
                test_preds += model.predict(X_test) / cv.n_splits

        # Select confident predictions
        confident_mask = (
            (test_preds >= self.confidence_threshold) |
            (test_preds <= 1 - self.confidence_threshold)
        )
```

```
import numpy as np
import pandas as pd
from sklearn.model_selection import StratifiedKFold

class PseudoLabeler:
    """
    Safe pseudo-labeling with confidence thresholding.

    Critical: Use OOF predictions to generate pseudo-labels,
    never use test predictions directly in training.
    """

    def __init__(self, model, confidence_threshold=0.9, max_iterations=3):
        self.model = model
        self.confidence_threshold = confidence_threshold
        self.max_iterations = max_iterations
        self.pseudo_label_history = []

    def generate_pseudo_labels(self, X_train, y_train, X_test, cv):
        """
        Generate pseudo-labels using OOF predictions.
        """
        # Generate OOF predictions on test set
        test_preds = np.zeros(len(X_test))

        for train_idx, val_idx in cv.split(X_train, y_train):
            X_tr, X_val = X_train.iloc[train_idx], X_train.iloc[val_idx]
            y_tr = y_train.iloc[train_idx]

            model = clone(self.model)
            model.fit(X_tr, y_tr)

            # Predict on test
            if hasattr(model, 'predict_proba'):
                test_preds += model.predict_proba(X_test)[:, 1] / cv.n_splits
            else:
                test_preds += model.predict(X_test) / cv.n_splits

        # Select confident predictions
        confident_mask = (
            (test_preds >= self.confidence_threshold) |
            (test_preds <= 1 - self.confidence_threshold)
        )

        pseudo_labels = (test_preds > 0.5).astype(int)
```

## 8.2 Curriculum Training

```
import numpy as np

class CurriculumTrainer:
    """
    Curriculum learning: train on easy examples first,
    gradually increase difficulty.
    """

    def __init__(self, model, difficulty_metric='prediction_confidence'):
        self.model = model
        self.difficulty_metric = difficulty_metric

    def calculate_difficulty(self, X, y, model=None):
        """
        Calculate difficulty scores for each sample.
        """
        if model is None:
            model = self.model

        if self.difficulty_metric == 'prediction_confidence':
            # Higher confidence = easier sample
            if hasattr(model, 'predict_proba'):
                probs = model.predict_proba(X)
                confidence = np.max(probs, axis=1)
                return 1 - confidence  # Lower confidence = harder
            else:
                return np.random.random(len(X))  # Fallback

        elif self.difficulty_metric == 'loss':
            # Higher loss = harder sample
            from sklearn.metrics import log_loss
            if hasattr(model, 'predict_proba'):
                probs = model.predict_proba(X)
                losses = [log_loss([y[i]], [probs[i]], labels=np.unique(y))
                         for i in range(len(y))]
                return np.array(losses)

        return np.random.random(len(X))

    def train_curriculum(self, X, y, n_stages=3, stage_ratio=0.5):
        """
        Train with curriculum learning.
        """
```

```
import numpy as np

class CurriculumTrainer:
    """
    Curriculum learning: train on easy examples first,
    gradually increase difficulty.
    """

    def __init__(self, model, difficulty_metric='prediction_confidence'):
        self.model = model
        self.difficulty_metric = difficulty_metric

    def calculate_difficulty(self, X, y, model=None):
        """
        Calculate difficulty scores for each sample.
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        if model is None:
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                probs = model.predict_proba(X)
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        elif self.difficulty_metric == 'loss':
            # Higher loss = harder sample
            from sklearn.metrics import log_loss
            if hasattr(model, 'predict_proba'):
                probs = model.predict_proba(X)
                losses = [log_loss([y[i]], [probs[i]], labels=np.unique(y))
                         for i in range(len(y))]
                return np.array(losses)

        return np.random.random(len(X))

    def train_curriculum(self, X, y, n_stages=3, stage_ratio=0.5):
        """
        Train with curriculum learning.
        """
        X_array = np.array(X) if isinstance(X, pd.DataFrame) else X
        y_array = np.array(y)
```

## 8.3 Distillation

```
import numpy as np
from sklearn.base import clone

class ModelDistiller:
    """
    Distill ensemble or large model into smaller deployable model.

    Key for production: large ensemble -> single fast model.
    """

    def __init__(self, teacher_models, student_model, temperature=3.0):
        self.teacher_models = teacher_models
        self.student_model = student_model
        self.temperature = temperature

    def generate_teacher_predictions(self, X):
        """
        Generate soft predictions from teacher ensemble.
        """
        predictions = []

        for model in self.teacher_models:
            if hasattr(model, 'predict_proba'):
                probs = model.predict_proba(X)
                predictions.append(probs)
            else:
                preds = model.predict(X)
                # Convert to soft predictions
                probs = np.zeros((len(preds), 2))
                probs[:, 0] = 1 - preds
                probs[:, 1] = preds
                predictions.append(probs)

        # Average teacher predictions
        ensemble_probs = np.mean(predictions, axis=0)

        # Apply temperature scaling
        soft_probs = self._temperature_scale(ensemble_probs, self.temperature)

        return soft_probs

    def _temperature_scale(self, probs, temperature):
        """
```

```
import numpy as np
from sklearn.base import clone

class ModelDistiller:
    """
    Distill ensemble or large model into smaller deployable model.

    Key for production: large ensemble -> single fast model.
    """

    def __init__(self, teacher_models, student_model, temperature=3.0):
        self.teacher_models = teacher_models
        self.student_model = student_model
        self.temperature = temperature

    def generate_teacher_predictions(self, X):
        """
        Generate soft predictions from teacher ensemble.
        """
        predictions = []

        for model in self.teacher_models:
            if hasattr(model, 'predict_proba'):
                probs = model.predict_proba(X)
                predictions.append(probs)
            else:
                preds = model.predict(X)
                # Convert to soft predictions
                probs = np.zeros((len(preds), 2))
                probs[:, 0] = 1 - preds
                probs[:, 1] = preds
                predictions.append(probs)

        # Average teacher predictions
        ensemble_probs = np.mean(predictions, axis=0)

        # Apply temperature scaling
        soft_probs = self._temperature_scale(ensemble_probs, self.temperature)

        return soft_probs

    def _temperature_scale(self, probs, temperature):
        """
        Apply temperature scaling to soften probabilities.
        """
        # Convert to logits
```

**Sources & Latest Evidence**

- NVIDIA: Pseudo-Labeling Guide (2025-09-18): developer.nvidia.com
- BirdCLEF 2024 Winner: Pseudo-Labeling (2024): kaggle.com/competitions/birdclef-2024

# PART 9 — Kaggle vs Production Reality

> ***TL;DR:*** *The gap between competition success and production impact is real. Production systems must handle continuous data flows, maintain low latency, provide interpretability, and operate within cost constraints. Understanding these differences early prevents architectural debt and deployment failures.*

**Key Points:**

- Production models face distribution drift; monitoring is non-negotiable
- Latency constraints often require model distillation or simplification
- Feature stores bridge the gap between offline training and online serving
- Cost optimization: inference cost often exceeds training cost in production

## 9.1 Scaling Challenges

### Latency vs Accuracy Tradeoffs

**Table 7 Production Latency Requirements by Use Case**

| Use Case | Max Latency | Typical Model | Competition Equivalent |
|---|---|---|---|
| Real-time bidding | < 10ms | Linear model, small GBDT | Single LGBM (100 trees) |
| Search ranking | < 50ms | Medium GBDT, small NN | 3-5 model ensemble |
| Recommendation | < 200ms | Large GBDT, two-tower NN | 10-20 model ensemble |
| Batch prediction | Minutes | Full ensemble, large NN | Competition stack |

## Model Compression Techniques

```
# Model compression for production deployment

class ModelCompressor:
    """
    Compress competition models for production deployment.
    """

    @staticmethod
    def quantize_model(model, quantize_type='float16'):
        """
        Quantize model weights for faster inference.
        """
        if hasattr(model, 'convert_to_tensorflow'):
            # TensorFlow Lite quantization
            import tensorflow as tf
            converter = tf.lite.TFLiteConverter.from_keras_model(model)
            converter.optimizations = [tf.lite.Optimize.DEFAULT]

            if quantize_type == 'int8':
                converter.target_spec.supported_types = [tf.int8]

            quantized_model = converter.convert()
            return quantized_model

        return model

    @staticmethod
    def prune_trees(model, n_trees=None):
        """
        Reduce ensemble size by selecting best trees.
        """
        if hasattr(model, 'estimators_'):
            # Random Forest / GBDT
            if n_trees and len(model.estimators_) > n_trees:
                # Select trees with best validation performance
                model.estimators_ = model.estimators_[:n_trees]

        return model

    @staticmethod
    def distill_ensemble(ensemble_models, X_train, y_train, student_model):
        """
        Distill ensemble into single model.
        """
```

```
# Model compression for production deployment

class ModelCompressor:
    """
    Compress competition models for production deployment.
    """

    @staticmethod
    def quantize_model(model, quantize_type='float16'):
        """
        Quantize model weights for faster inference.
        """
        if hasattr(model, 'convert_to_tensorflow'):
            # TensorFlow Lite quantization
            import tensorflow as tf
            converter = tf.lite.TFLiteConverter.from_keras_model(model)
            converter.optimizations = [tf.lite.Optimize.DEFAULT]

            if quantize_type == 'int8':
                converter.target_spec.supported_types = [tf.int8]

            quantized_model = converter.convert()
            return quantized_model

        return model

    @staticmethod
    def prune_trees(model, n_trees=None):
        """
        Reduce ensemble size by selecting best trees.
        """
        if hasattr(model, 'estimators_'):
            # Random Forest / GBDT
            if n_trees and len(model.estimators_) > n_trees:
                # Select trees with best validation performance
                model.estimators_ = model.estimators_[:n_trees]

        return model

    @staticmethod
    def distill_ensemble(ensemble_models, X_train, y_train, student_model):
        """
        Distill ensemble into single model.
        """
        # Generate ensemble predictions
        ensemble_preds = []
```

## 9.2 Monitoring & MLOps Gaps

## Production Monitoring Framework

```
import pandas as pd
import numpy as np
from datetime import datetime, timedelta

class ProductionMonitor:
    """
    Monitor production model performance and data drift.
    """

    def __init__(self, model, reference_data, reference_predictions):
        self.model = model
        self.reference_data = reference_data
        self.reference_preds = reference_predictions
        self.monitoring_history = []

    def check_prediction_drift(self, new_predictions, threshold=0.1):
        """
        Detect drift in prediction distribution.
        """
        from scipy.stats import ks_2samp

        statistic, pvalue = ks_2samp(self.reference_preds, new_predictions)

        drift_detected = pvalue < threshold

        return {
            'ks_statistic': statistic,
            'pvalue': pvalue,
            'drift_detected': drift_detected,
            'reference_mean': np.mean(self.reference_preds),
            'current_mean': np.mean(new_predictions),
        }

    def check_feature_drift(self, new_data, features, threshold=0.05):
        """
        Detect drift in feature distributions.
        """
        from scipy.stats import ks_2samp

        drift_results = {}

        for feature in features:
            if feature in self.reference_data.columns:
                ref_values = self.reference_data[feature].dropna()
```

```
import pandas as pd
import numpy as np
from datetime import datetime, timedelta

class ProductionMonitor:
    """
    Monitor production model performance and data drift.
    """

    def __init__(self, model, reference_data, reference_predictions):
        self.model = model
        self.reference_data = reference_data
        self.reference_preds = reference_predictions
        self.monitoring_history = []

    def check_prediction_drift(self, new_predictions, threshold=0.1):
        """
        Detect drift in prediction distribution.
        """
        from scipy.stats import ks_2samp

        statistic, pvalue = ks_2samp(self.reference_preds, new_predictions)

        drift_detected = pvalue < threshold

        return {
            'ks_statistic': statistic,
            'pvalue': pvalue,
            'drift_detected': drift_detected,
            'reference_mean': np.mean(self.reference_preds),
            'current_mean': np.mean(new_predictions),
        }

    def check_feature_drift(self, new_data, features, threshold=0.05):
        """
        Detect drift in feature distributions.
        """
        from scipy.stats import ks_2samp

        drift_results = {}

        for feature in features:
            if feature in self.reference_data.columns:
                ref_values = self.reference_data[feature].dropna()
                new_values = new_data[feature].dropna()
```

## MLOps Integration Checklist

### Production Deployment Checklist

- ☐ Model versioned and registered in MLflow/model registry
- ☐ Feature pipeline tested for online/offline consistency
- ☐ A/B testing framework configured
- ☐ Monitoring dashboards created (latency, throughput, errors)
- ☐ Drift detection configured with alert thresholds
- ☐ Rollback procedure documented and tested
- ☐ Cost monitoring enabled
- ☐ Documentation updated (model card, API spec)

### Sources & Latest Evidence

- MLOps Best Practices 2024: github.com/umitkacar/awesome-mlops
- MLflow Documentation: mlflow.org/docs
- Evidently AI: ML Monitoring: evidentlyai.com

# PART 10 — Reverse Engineering Wins

> ***TL;DR:*** *Analyzing winning solutions reveals patterns that generalize across competitions. The 2024 landscape shows ensembles of 50-100+ models, sophisticated feature engineering, and GPU-accelerated experimentation as common themes among top finishers.*

**Key Points:**

- AutoGluon dominated 2024 tabular competitions (15 of 18)
- Winning ensembles typically include LGBM, XGB, CAT, and NNs
- Feature engineering remains the highest-ROI activity
- GPU acceleration (RAPIDS) is now standard for top competitors

## 10.1 Annotated Winner Breakdowns

### Case Study: Podcast Listening Time (April 2025)

**Winner:** Chris Deotte (Kaggle Grandmaster)

**Approach:** Three-level stack of 72 models using RAPIDS cuML on GPU

**Table 8 Podcast Competition Winner Architecture**

| Level | Models | Key Insight |
|---|---|---|
| Level 0 | XGB, LGBM, CAT, NN, TabPFN, KNN, SVR, Ridge, RF | Maximum diversity across algorithms |
| Level 1 | Ridge regression on L0 OOF predictions | Learn optimal model combinations |

| | | |
|---|---|---|
| Level 2 | Gradient boosting on L1 predictions | Final refinement |

**Key Techniques:**

- Treating every number as both numeric and categorical
- Thousands of groupby features
- GPU acceleration for fast experimentation
- Hill climbing for ensemble weight optimization

### Case Study: Insurance Prediction (December 2024)

**Winner:** Single XGBoost model with 611 features

**Approach:** Feature engineering focus rather than massive ensembles

```
# Key feature engineering techniques from winner:

# 1. Multiple encodings for each categorical
encodings = ['label', 'target_mean', 'target_median', 'count', 'nunique']

# 2. Groupby statistics at multiple levels
groupby_levels = [
    ['category_1'],
    ['category_2'],
    ['category_1', 'category_2'],
]

# 3. Interaction features
interactions = [
    ('num_1', 'num_2', 'multiply'),
    ('num_1', 'num_2', 'divide'),
    ('cat_1', 'num_1', 'groupby_mean'),
]

# 4. Target encoding with high smoothing
smoothing = 10.0  # Higher than typical
```

## 10.2 What Generalizes

## Patterns from 2024 Winners

**Table 9 Generalizable Patterns from 2024 Winners**

| Pattern | Frequency | Transfer to Production |
|---|---|---|
| GBDT ensemble (3-5 models) | 95% | Direct (with distillation) |
| Groupby aggregations | 100% | Direct (feature store) |
| Target encoding | 90% | Direct (with monitoring) |
| Deep stacking (3+ levels) | 40% | Limited (complexity) |
| Neural networks | 60% | Direct (with optimization) |
| GPU acceleration | 80% | Direct (cost permitting) |

## Anti-Patterns to Avoid

### Competition Anti-Patterns

- **LB overfitting:** Tuning to public leaderboard
- **Leakage exploitation:** Using future information
- **Ensemble bloat:** 100+ similar models
- **Feature explosion:** 10K+ features without selection
- **Complexity for complexity's sake:** 5-level stacks

### Sources & Latest Evidence

- Podcast Competition Winner (2025-04): kaggle.com/competitions/playground-series-s5e4
- Insurance Competition Winner (2024-12): kaggle.com/competitions/playground-series-s4e12
- State of ML Competitions 2024 (2025-02-25): mlcontests.com

# PART 11 — Career & Strategic Leverage

> ***TL;DR:*** *Kaggle success translates to career opportunities when positioned correctly. Hiring managers value the demonstrated ability to solve complex problems, write clean code, and deliver results under pressure. The key is articulating competition skills in production-relevant terms.*

**Key Points:**

- Kaggle rankings signal problem-solving ability and persistence
- Portfolio artifacts (clean code, write-ups) matter more than medals
- Grandmaster status opens doors; Expert/Contributor shows growth trajectory
- Team competitions demonstrate collaboration skills

## 11.1 Hiring Manager POV

### What Hiring Managers Look For

**Table 10 Kaggle Signals for Hiring**

| Signal | What It Demonstrates | Relevance |
|---|---|---|
| Grandmaster | World-class ML expertise, consistency | Very High |
| Master + medals | Strong practitioner, proven results | High |
| Detailed write-ups | Communication, knowledge sharing | High |
| Clean code repos | Engineering discipline | High |
| Expert rank | Growth mindset, learning ability | Medium |

| | | |
|---|---|---|
| Competition count | Persistence, iteration speed | Medium |

## Positioning Your Kaggle Experience

```
# Resume bullet templates for Kaggle experience

# For Grandmasters:
"Achieved Kaggle Grandmaster status (top 0.1% of 200K+ data scientists)
with 5 gold medals in competitions including [specific domains].
Consistently delivered top-1% solutions through systematic experimentation
and ensemble optimization."

# For Masters:
"Kaggle Master with 3 competition medals. Developed production-ready
ML pipelines achieving top-5% rankings in [domain] competitions.
Strong expertise in feature engineering and model ensembling."

# For Experts/Contributors:
"Active Kaggle competitor with focus on [domain]. Implemented
state-of-the-art techniques including [specific methods].
Published detailed solution write-ups with reproducible code."

# Interview talking points:
TALKING_POINTS = {
    'problem_solving': [
        'Approached novel problems with systematic methodology',
        'Iterated rapidly through 50+ experiments per competition',
        'Balanced exploration vs exploitation under time constraints',
    ],
    'technical_skills': [
        'Built end-to-end ML pipelines from EDA to deployment',
        'Optimized models for both accuracy and efficiency',
        'Implemented custom architectures for specific problem types',
    ],
    'collaboration': [
        'Participated in team competitions with distributed members',
        'Shared knowledge through public kernels and discussions',
        'Reviewed and integrated feedback from community',
    ],
}
```

## 11.2 Portfolio Artifacts

### Essential Portfolio Components

**Kaggle Portfolio Checklist**

- ☐ GitHub repo with clean, documented code for top solutions
- ☐ Detailed write-ups on competition approaches (blog/Kaggle)
- ☐ Public kernels demonstrating specific techniques
- ☐ Dataset contributions or feature analyses
- ☐ Discussion participation showing thought leadership

### GitHub Repository Template

```
kaggle-portfolio/
├── README.md                    # Overview of competitions and results
├── COMPETITIONS.md             # Detailed competition history
├── techniques/                 # Reusable technique implementations
│   ├── feature_engineering/
│   ├── ensemble_methods/
│   └── model_architectures/
├── competitions/               # Competition-specific code
│   ├── competition_1/
│   │   ├── README.md
│   │   ├── notebook.ipynb
│   │   └── src/
│   └── competition_2/
└── utils/                      # Shared utilities
    ├── cv_utils.py
    ├── metrics.py
    └── plotting.py
```

## Write-up Structure

```
# Competition Write-up Template

TITLE: "[Competition Name] - [Rank] Place Solution"

SECTIONS:
1. Executive Summary
   - Final rank and score
   - High-level approach
   - Key differentiators

2. Problem Analysis
   - Data characteristics
   - Evaluation metric
   - Key challenges

3. Solution Overview
   - Pipeline diagram
   - Major components

4. Feature Engineering
   - Key features created
   - Encoding strategies
   - Feature selection

5. Model Architecture
   - Base models used
   - Ensemble strategy
   - Hyperparameters

6. Validation Strategy
   - CV approach
   - CV-LB correlation
   - Leakage prevention

7. Key Insights
   - What worked
   - What didn't
   - Lessons learned

8. Code & Reproducibility
   - GitHub link
```

```
# Competition Write-up Template

TITLE: "[Competition Name] - [Rank] Place Solution"

SECTIONS:
1. Executive Summary
   - Final rank and score
   - High-level approach
   - Key differentiators

2. Problem Analysis
   - Data characteristics
   - Evaluation metric
   - Key challenges

3. Solution Overview
   - Pipeline diagram
   - Major components

4. Feature Engineering
   - Key features created
   - Encoding strategies
   - Feature selection

5. Model Architecture
   - Base models used
   - Ensemble strategy
   - Hyperparameters

6. Validation Strategy
   - CV approach
   - CV-LB correlation
   - Leakage prevention

7. Key Insights
   - What worked
   - What didn't
   - Lessons learned

8. Code & Reproducibility
   - GitHub link
   - Dependencies
   - Run instructions
```

**Sources & Latest Evidence**

- Kaggle Careers Page: kaggle.com/careers
- ML Engineer Hiring Trends (2024): linkedin.com

# PART 12 — Systems Toolkit & 90-Day Mastery Plan

> ***TL;DR:*** *Mastery comes from deliberate practice with structured goals. This 90-day plan provides week-by-week milestones, checklists, and a complete repository structure to accelerate your journey from competent practitioner to competitive data scientist.*

**Key Points:**

- Week 1-2: Master CV architecture and baseline models
- Week 3-6: Intensive feature engineering practice
- Week 7-10: Ensemble methods and stacking
- Week 11-12: Specialization and competition execution

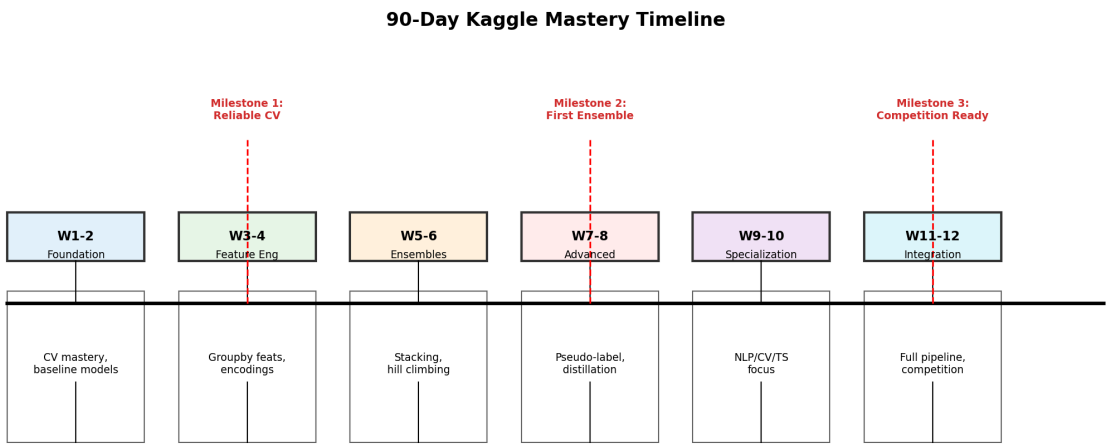


**Figure 7:** 90-Day Kaggle Mastery Timeline

## Competition Start Checklist

### Day 1-2: Setup & EDA

- ☐ Read competition description and evaluation metric
- ☐ Download and verify data integrity
- ☐ Run leakage detection suite
- ☐ Create initial EDA notebook
- ☐ Identify data type (tabular, NLP, CV, time series)
- ☐ Set up experiment tracking
- ☐ Establish baseline model

### Week 1: Validation Strategy

- ☐ Implement appropriate CV strategy
- ☐ Validate CV-LB correlation with early submissions
- ☐ Test 2-3 different model families
- ☐ Document baseline performance
- ☐ Identify key features through importance analysis

### Feature Engineering Checklist

- ☐ Create groupby aggregations at multiple levels
- ☐ Implement safe target encoding
- ☐ Generate interaction features
- ☐ Add date/time features if applicable
- ☐ Test polynomial features for numerical data
- ☐ Apply feature selection
- ☐ Validate each feature group with CV

### Ensemble Checklist

- ☐ Train diverse base models (LGBM, XGB, CAT, NN)
- ☐ Measure prediction diversity (correlation matrix)

- ☐ Generate OOF predictions for all models
- ☐ Implement two-level stacking
- ☐ Optimize weights with hill climbing
- ☐ Test different meta-learners
- ☐ Validate ensemble doesn't overfit

## 12.2 90-Day Plan

### Phase 1: Foundation (Weeks 1-2)

**Table 11 90-Day Mastery Plan - Phase 1**

| Week | Focus | Deliverables | Success Metric |
|---|---|---|---|
| 1 | CV Mastery | Implement 3 CV strategies | CV-LB correlation > 0.9 |
| 2 | Baseline Models | LGBM, XGB, CAT baselines | Top 50% on playground |

### Phase 2: Feature Engineering (Weeks 3-6)

**Table 12 90-Day Mastery Plan - Phase 2**

| Week | Focus | Deliverables | Success Metric |
|---|---|---|---|
| 3 | Groupby Features | 100+ groupby features | +2% CV improvement |
| 4 | Target Encoding | Safe TE implementation | +1% CV improvement |
| 5 | Interactions | Polynomial & cat interactions | +1% CV improvement |
| 6 | Feature Selection | Selection pipeline | Reduced feature count, maintained performance |

### Phase 3: Ensembling (Weeks 7-10)

**Table 13 90-Day Mastery Plan - Phase 3**

| Week | Focus | Deliverables | Success Metric |
|---|---|---|---|
| 7 | Model Diversity | 5+ diverse base models | Correlation matrix < 0.9 |
| 8 | Stacking | 2-level stack implementation | +2% over best single |
| 9 | Blending | Hill climbing optimizer | Optimal weights found |
| 10 | Advanced | Pseudo-labeling pipeline | +1% improvement |

### Phase 4: Integration (Weeks 11-12)

**Table 14 90-Day Mastery Plan - Phase 4**

| Week | Focus | Deliverables | Success Metric |
|---|---|---|---|
| 11 | Specialization | NLP/CV/TS focus area | Top 25% in specialty |
| 12 | Competition | Full competition entry | Medal or top 10% |

## 12.3 Repo Structure

## Complete Repository Template

```
kaggle-master/
├── README.md
├── requirements.txt
├── setup.py
├── .gitignore
├── config/
│   ├── base.yaml
│   ├── cv_strategies.yaml
│   └── model_configs/
├── data/
│   ├── raw/
│   ├── processed/
│   └── external/
├── notebooks/
│   ├── template_eda.ipynb
│   ├── template_baseline.ipynb
│   └── template_submission.ipynb
├── src/
│   ├── __init__.py
│   ├── data/
│   │   ├── __init__.py
│   │   ├── load_data.py
│   │   ├── preprocess.py
│   │   ├── features.py
│   │   └── leakage_detector.py
│   ├── models/
│   │   ├── __init__.py
│   │   ├── train.py
│   │   ├── predict.py
│   │   ├── ensemble.py
│   │   └── stacking.py
│   ├── cv/
│   │   ├── __init__.py
│   │   ├── strategies.py
│   │   └── validator.py
│   └── utils/
│       ├── __init__.py
│       ├── metrics.py
│       ├── logging.py
│       └── visualization.py
├── experiments/
├── submissions/
├── models/
└── tests/
```

```
kaggle-master/
├── README.md
├── requirements.txt
├── setup.py
├── .gitignore
├── config/
│   ├── base.yaml
│   ├── cv_strategies.yaml
│   └── model_configs/
├── data/
│   ├── raw/
│   ├── processed/
│   └── external/
├── notebooks/
│   ├── template_eda.ipynb
│   ├── template_baseline.ipynb
│   └── template_submission.ipynb
├── src/
│   ├── __init__.py
│   ├── data/
│   │   ├── __init__.py
│   │   ├── load_data.py
│   │   ├── preprocess.py
│   │   ├── features.py
│   │   └── leakage_detector.py
│   ├── models/
│   │   ├── __init__.py
│   │   ├── train.py
│   │   ├── predict.py
│   │   ├── ensemble.py
│   │   └── stacking.py
│   ├── cv/
│   │   ├── __init__.py
│   │   ├── strategies.py
│   │   └── validator.py
│   └── utils/
│       ├── __init__.py
│       ├── metrics.py
│       ├── logging.py
│       └── visualization.py
├── experiments/
├── submissions/
├── models/
└── tests/
    ├── test_features.py
```

## Reproducibility Audit Checklist

### Pre-Submission Audit

- ☐ All random seeds set (numpy, random, torch, tensorflow)
- ☐ Dependencies pinned in requirements.txt
- ☐ Code runs end-to-end without errors
- ☐ CV results reproducible across runs
- ☐ OOF predictions saved and versioned
- ☐ Feature pipeline documented
- ☐ Model artifacts saved
- ☐ Experiment logged with all parameters

## Final Week Plan

```
# Final Week Execution Plan

DAY_MINUS_7:
- Freeze all new feature development
- Validate best 3 approaches with full CV
- Document all hyperparameters

DAY_MINUS_5:
- Generate predictions for all candidate models
- Create ensemble combinations
- Validate ensemble diversity

DAY_MINUS_3:
- Select final 2-3 submissions
- Prepare backup submissions
- Test submission pipeline end-to-end

DAY_MINUS_1:
- Submit primary entries
- Prepare emergency backup
- No new experiments

FINAL_HOURS:
- Monitor for any issues
- Submit backups if needed
- Document final approach
```

### Continuous Improvement Habits

- **Daily:** Review one competition solution write-up
- **Weekly:** Implement one new technique from research
- **Monthly:** Complete one full competition cycle
- **Quarterly:** Publish one detailed solution write-up

### Sources & Latest Evidence

- Kaggle Learn: kaggle.com/learn
- Kaggle Discussions: kaggle.com/discussions
- Winning Solutions Collection: kaggle.com/code/sudalairajkumar

# Closing Thoughts

This handbook represents a synthesis of patterns from 2024-2025 competition winners, production ML engineering practices, and the collective wisdom of the Kaggle community. The techniques and systems described here are not shortcuts—they are the result of disciplined experimentation, rigorous validation, and continuous learning.

The practitioners who succeed both on Kaggle and in production share a common trait: they treat machine learning as a systems engineering discipline. They understand that a 1% improvement from a better CV strategy is worth more than a 5% improvement that doesn't generalize. They value reproducibility over one-off results. They build pipelines that can be iterated, not notebooks that can only be run once.

Use this handbook as a reference, a starting point, and a sanity check. Adapt the patterns to your specific problems. Question the assumptions. And most importantly, build things that matter.

*Compiled by Amer Hussein, 2025*
*LinkedIn: linkedin.com/in/amer-hussein*